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Distinguishing Plume and Metasomatized Lithospheric Mantle Contributions to Post-Flood Basalt Volcanism on the Southeastern Ethiopian Plateau

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ABSTRACT

Magmatism in the East African Rift System (EARS) contains a spatial and temporal record of changing contributions from the Afar mantle plume, anciently metasomatized lithosphere, the upper mantle and the continental crust. A full understanding of this record requires characterizing volcanic products both within the rift valley and on its flanks. In this study, three suites of mafic, transitional to alkaline lavas, were collected over a northeast-southwest distance of ~150 km along the southeastern Ethiopian Plateau, adjacent to the Main Ethiopian Rift. Specifically, late Oligocene to Quaternary mafic lavas were collected from Chiro, Debre Sahil and the Bale Mountains. New major element, trace element, $^{40}Ar/^{39}Ar$ ages and isotopic results (Sr, Nd, Pb, Hf, Os, He) show spatial and temporal variation in the lavas caused by dynamical changes in the source of volcanism during the evolution of the EARS. The trace element compositions of Oligocene and Miocene Chiro lavas indicate derivation from mildly depleted and nominally anhydrous lithospheric mantle, with variable inputs from the crust. Further south, Miocene Debre Sahil and alkaline Bale Mountains lavas have enriched incompatible trace element ratios (e.g. Ba/Nb = 12–43, La/Sm_N = 3·1–4·9, Tb/Yb_N = 1·6–2·4). Additionally, their $^{87}Sr/^{86}Sr$, $^{143}Nd/^{144}Nd$, $^{176}Hf/^{177}Hf$ and $^{206}Pb/^{204}Pb$ values trend toward a radiogenic Pb (HIMU) component. Radiogenic $^{187}Os/^{188}Os$ in these lavas correlates positively with $^{206}Pb/^{204}Pb$ and trace element indicators consistent with ancient metasomatic enrichment of their mantle source. In contrast, transitional Miocene Bale Mountains lavas have lower incompatible trace element abundances, less enriched trace element ratios (Ba/Nb ~7, La/Sm_N = 2·3–2·5) and less radiogenic isotopic signatures that originate from melting garnet-bearing, anhydrous lithospheric mantle (Tb/Yb_N = 2·5–2·9). Pliocene and Quaternary Bale Mountains basaltic lavas are chemically and isotopically similar to Main Ethiopian Rift lavas. Trace element and isotopic indicators in both of these suites denote an amphibole-bearing source distinct from that sampled by the older Bale Mountains lavas. Isotopically, Pliocene and Quaternary Bale lavas have notably less radiogenic Sr–Nd–Pb–Hf isotopic ratios. Quaternary Bale Mountains lavas have the strongest mantle plume contribution ($^3He/^4He$ = 12·1–12·5 R_A), while other Bale Mountains, Debre Sahil and Chiro lavas were derived dominantly by melting of lithospheric or upper mantle sources ($^3He/^4He$ = 5·1–9·1 R_A). A multi-stage, regional-scale model of metasomatism and partial melting accounts

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for the spatial and temporal variations on the southeastern Ethiopian Plateau. Early Debre Sahil and alkaline Bale Mountains mafic lavas are melts derived from Pan-African lithosphere containing amphibole-bearing metasomes, while later transitional Bale basalts are melts of lithosphere containing anhydrous, clinopyroxene-rich veins. These ancient metasomatized domains were eventually removed through preferential melting, potentially during thermal erosion of the lithosphere or lithospheric foundering. Pliocene and Quaternary Bale Mountains lavas erupted after tectonic extension progressed throughout Ethiopia and was accompanied by increased plume influence on the volcanic products.

Key words: East African rift; mantle plume; metasomatism; HIMU; Ethiopian Plateau

INTRODUCTION

The East African Rift System (EARS) has been shaped over the past 45 Ma by magmatism and tectonic extension. The EARS is superimposed on a continental nucleus that was assembled during the Pan-African orogeny (1220 to 500 Ma; Jackson & Ramsay, 1980; Stein & Goldstein, 1996; Meert, 2003), with subduction-related magmatism occurring between 900 and 540 Ma (Bentor, 1985). Mantle xenoliths and mafic lavas from the African and Arabian plates contain a record of these ancient melt depletion events in their trace element and isotopic compositions. They also have sampled lithospheric domains that range from highly depleted to highly enriched, metasomatized regions (e.g. Lloyd, 1981; Dawson & Smith, 1988; Davies & Lloyd, 1989; Lloyd *et al.*, 1985, 1991; Stein & Goldstein, 1996; George *et al.*, 1998; Kaeser *et al.*, 2006, 2009; Koornneef *et al.*, 2009; Rosenthal *et al.*, 2009; Rooney *et al.*, 2014; Sgualdo *et al.*, 2015). Because mafic lavas are far more widespread in their occurrence than xenoliths, they arguably provide the best means for reconstructing the temporal evolution of the EARS and for understanding how magmatism varied in its sampling of distinct mantle domains (e.g. mantle plume, the lithosphere and convecting asthenosphere). A major challenge has been discriminating between the influence of Pan-African and earlier geochemical signatures trapped in the continental lithosphere (e.g. Stein *et al.*, 1997; Reisberg *et al.*, 2004) from geochemical signatures related to younger mantle plume activity and extensional magmatism (e.g. Schilling *et al.*, 1992; Hofmann *et al.*, 1997; Späth *et al.*, 2001; Rooney *et al.*, 2012).

Here we investigate geochemical variations in mafic alkaline lavas erupted in three areas on the southeast portion of the Ethiopia Plateau, following the major pulse of flood basalt volcanism at ~30 Ma (Fig. 1). Geochemical variations in lavas from this southeastern sector of the Plateau contrast markedly with the earlier flood basalts and the subsequent rift-related basalts. Their geochemical variations help identify how the sources of volcanism in this region away from the main axis of rifting have varied temporally and spatially. We present $^{40}Ar/^{39}Ar$ age results, plus major, trace element and Pb-Nd-Hf-Sr-Os-He isotope measurements that document how basalt compositions vary as a function of age and distance from the central region of the Plateau. Our three study localities, Chiro (formerly known as Asbe Teferi), Debre Sahil and the Bale Mountains (Bale), provide stratigraphic sequences of lavas that record very late-stage flood basalt activity, Miocene shield volcanism and young basalts that erupted contemporaneously with Main Ethiopian Rift (MER) basalts. Collectively these three sample suites provide an integrated view of lithospheric evolution and the changing sources of magmatism during plume-related rifting in the EARS.

GEOLOGIC SETTING

Overview of the East African rift system

The EARS has been the site of volcanic activity since the middle Eocene, beginning with moderate volumes of mafic lavas in southern Ethiopia and northern Kenya between 45–34 Ma and followed by ~30 Ma Afar plume volcanism further north (George *et al.*, 1998; George & Rogers, 2002; Furman *et al.*, 2006b; Rooney, 2017). This later phase of activity was associated with the break-up

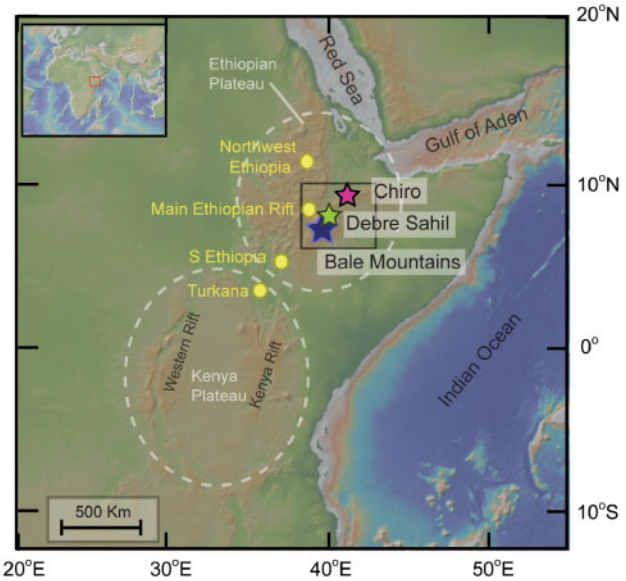


Fig. 1. Index map of the EARS showing sample areas on the southeastern portion of the Ethiopian Plateau. The square illustrates the location of Figure 2. The basemap was generated using GeoMapApp.

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of the Afro-Arabian shield, as the arrival of the Afar plume head at ~30 Ma initiated rifting in the Red Sea (~28 Ma; e.g. Hofmann *et al.*, 1997; Wolfenden *et al.*, 2005) and the Gulf of Aden (~27 Ma; Fantozzi, 1996). A concurrent pulse of volcanism ~25 Ma in the Turkana region of northern Kenya reactivated Cenozoic Anza graben faults and generated new ruptures (Morley *et al.*, 1992; Hendrie *et al.*, 1994; Furman *et al.*, 2006b). Volcanism and extension propagated both to the north and the south: extension to the north created the Main Ethiopian Rift (MER) and reached the Afar triple junction at around 11 Ma (e.g. Wolfenden *et al.*, 2004); southward extension propagated around the Tanzanian craton, producing the Western and Eastern Rifts as early as 22 Ma (Roberts *et al.*, 2012) and reaching southern Tanzania at ~1 Ma (e.g. Ebinger *et al.*, 1989).

Prior investigations of volcanism in Ethiopia have focused on the 350 000 km^3 of ~30 Ma plateau flood basalts (e.g. Baker *et al.*, 1996a; Hofmann *et al.*, 1997; Pik *et al.*, 1998, 1999; Kieffer *et al.*, 2004; Furman *et al.*, 2006a; Meshesha & Shinjo, 2007; Beccaluva *et al.*, 2009; Rogers *et al.*, 2010). These mafic lavas have been classified on the basis of their TiO_2 contents as forming three groups, two with high levels of TiO_2 (HT1 and HT2) and one with lower TiO_2 (LT); HT2 basalts and picrites from northwest Ethiopia were thought to reflect most closely the Afar plume-head composition (Pik *et al.*, 1998, 1999; Kieffer *et al.*, 2004; Beccaluva *et al.*, 2009; Rogers *et al.*, 2010). More recent work suggests the HT2 flood basalts also contain significant contributions from garnet-bearing pyroxenite or eclogite (Furman *et al.*, 2016; Natali *et al.*, 2016). LT lavas were derived from a more depleted mantle than HT2 lavas and may originate from the lithospheric mantle, a compositionally distinct portion of the Afar plume, or plume-metasomatized asthenospheric mantle (Pik *et al.*, 1998, 1999; Beccaluva *et al.*, 2009). The origin of LT lavas is ambiguous due to large crustal residence times (Krans *et al.*, 2018). HT1 lavas occupy a compositional space between HT2 and LT lavas (Pik *et al.*, 1999). Small volcanic provinces and shield volcanoes formed to the south of the Plateau both before (45–35 Ma) and after (28–10 Ma) peak plume activity; the largest volume of volcanic material is found on the northwestern Ethiopian Plateau (Berhe *et al.*, 1987; George *et al.*, 1998; George & Rogers, 2002; Ukstins *et al.*, 2002; Coulie *et al.*, 2003; Kieffer *et al.*, 2004; Beccaluva *et al.*, 2009). Today, the flood basalt province is dissected by the MER with its associated mafic volcanic rocks (Hart *et al.*, 1989; Chernet *et al.*, 1998, Chernet & Hart, 1999; Trua *et al.*, 1999; Peccerillo *et al.*, 2003; Furman *et al.*, 2006a; Rooney *et al.*, 2007, 2012). Recent lavas from Erta 'Ale (Deniel *et al.*, 1994; Barrat *et al.*, 1998) are considered to be chemically representative of the modern Afar plume tail (Furman *et al.*, 2006a).

The existence of one or more mantle plumes has been proposed throughout the EARS (e.g. Pik *et al.*, 1999; George & Rogers, 2002; Furman *et al.*, 2006b; Rooney *et al.*, 2012; Castillo *et al.*, 2014; Rooney, 2017). High $^3He/^4He$ ratios (9–19·6 R_A) and Sr–Nd–Pb isotopic compositions associated with flood basalt and rift-related volcanism are attributed consistently to the Afar mantle plume (Marty *et al.*, 1996; Pik *et al.*, 2006; Hilton *et al.*, 2011; Rooney *et al.*, 2012; Halldórsson *et al.*, 2014). It is worth noting that most pre- and post-flood basalt mafic lavas do not resemble the ~30 Ma flood basalts geochemically or isotopically, though some late Miocene to recent lavas do show elevated $^3He/^4He$ ratios (Hilton *et al.*, 2011; Rooney *et al.*, 2012). Lavas erupted during the late Oligocene through mid-Miocene across Ethiopia and northern Kenya are more alkaline and, more often than not, trend isotopically toward a HIMU component (e.g. George & Rogers, 2002; Kieffer *et al.*, 2004; Furman *et al.*, 2006b; Rooney *et al.*, 2014) that is distinct from the inferred Afar plume composition (Rooney *et al.*, 2012). The widespread occurrence and varying intensity of this geochemical signature in late Oligocene to mid-Miocene mafic volcanism suggests that it is present to some degree beneath a significant portion of the EARS. Notably, no HIMU-trending isotopic ratios are observed in young (<10 Ma) lavas.

Field description and petrographic overview

The study region is represented on two geological maps (Kazmin, 1973; Merla *et al.*, 1979) that differ in their approach to stratigraphic classification. The geologic map of Merla *et al.* (1979) groups the lower mafic units in all three sample localities as a single upper Oligocene to upper Miocene unit referred to as the Arussi and Bale basalts; here, we follow the more detailed classification scheme of Kazmin (1973). Samples were collected along a northeast–southwest transect near the towns of Chiro and Debre Sahil and throughout the Bale Mountains (Fig. 2).

Chiro

Chiro is located on the Plateau just east of where the MER widens into the Afar Depression (Fig. 2). The surrounding region is marked by abundant Oligocene–Miocene alkali olivine basalts and tuffs with sparse rhyolites; this is mapped broadly as part of the Ashangi Group trap series and dissected by numerous faults. This unit is overlain by the smaller volume Magdala group with upper Miocene to Pleistocene rhyolites, trachytes, trachytic tuffs, ignimbrite agglomerates and basalt (Kazmin, 1973).

The basal basaltic layer (CH-02) in contact with Mesozoic sandstones is a massive aphyric basalt with sparse vesicles and is among the oldest lavas reported in the region (e.g. Berhe *et al.*, 1987). Overlying this unit, the volcanic deposits consist of a series of aphyric to mildly aphanitic mafic lavas interlayered with more felsic tuff deposits. Near the town of Chiro, a tuff unit (CH-14) caps the volcanic sequence, while further to the southwest the capping unit consists of moderately vesicular basalt (CH-12).

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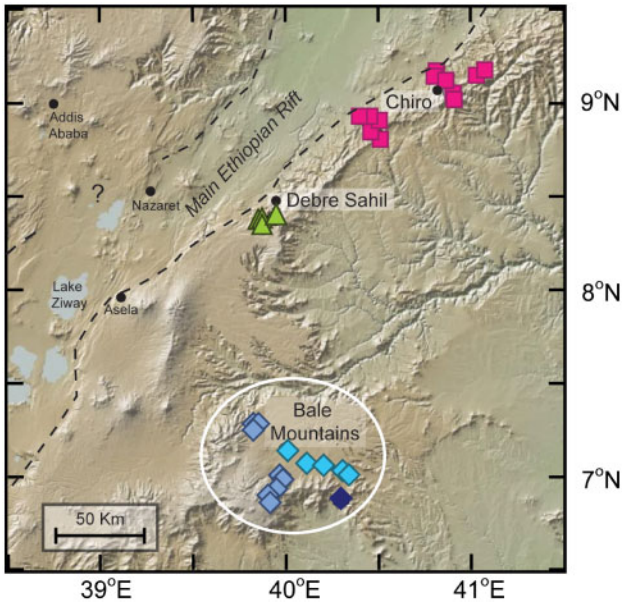


Fig. 2. Map showing sample locations near the towns of Chiro and Debre Sahil and throughout the Bale Mountains region (circled in white). The Miocene Bale lavas (dark blue diamonds) were sampled over a vertical cross section. Consequently, the samples appear as a single symbol on the map.

Mafic lavas from Chiro are dominantly aphyric to aphanitic, mildly vesicular with rare plagioclase and clinopyroxene phenocrysts. The groundmass consists mostly of plagioclase laths and microcrystals of clinopyroxene, olivine and Fe–Ti oxides with varying amounts of basaltic glass. All samples have experienced some degree of alteration, including vesicles partially filled with alteration products. Two Chiro samples, CH-05 and CH-09, are petrographically distinct: basal unit CH-05 is an olivine-bearing, fine-grained, slightly altered mafic lava with $<$ 1% plagioclase phenocrysts. Sample CH-09 from the middle of the sequence contains ~25% subhedral to anhedral olivine phenocrysts up to ~6 mm long, accompanied by plagioclase feldspar phenocrysts $\leq$ 4 mm. The high olivine modal abundance suggests the olivine was accumulated prior to eruption.

Debre Sahil

Volcanism near the town of Debre Sahil is mapped as part of the Ashangi Group trap series (Kazmin, 1973), but vegetation and limited accessibility make stratigraphic correlation difficult (Fig. 3) and obscure the contact between volcanic rocks and underlying Mesozoic sediments. We estimate the volcanic pile to be 500–700 m thick. Petrographically, Debre Sahil lavas are dominantly porphyritic with olivine, plagioclase and clinopyroxene phenocrysts. Most samples contain groundmass plagioclase, clinopyroxene, olivine and Fe–Ti oxides, with varying amounts of devitrified basaltic glass. Olivine and some clinopyroxene phenocrysts display alteration along rims and fractures.

Bale Mountains

Bale Mountains mafic lavas are part of the Ashangi Group trap series (Kazmin, 1973) and are associated with younger lavas referred to as Pleistocene to Present Afar lavas (also mapped within the nearby rift valley, Merla *et al.*, 1979) or as Miocene Shield Group lavas of the Trap Series (Kazmim, 1973). We sampled stratigraphic sequences through two key eruptive periods: (1) Late Oligocene to early Miocene Bale lavas include the Lower Stratoid flood basalts (Merla *et al.*, 1979; Berhe *et al.*, 1987); sample BM-12 lies uncomformably on Mesozoic sandstones (Fig. 3). Above this unit, basaltic and trachytic materials alternate, with distinctive red paleosols between samples BM-18 and BM-19. (2) Pliocene and Quaternary Bale lavas overlap with a suite of samples collected from Mt. Batu. Berhe *et al.* (1987) mapped these as Pliocene Sanete basalts and Quaternary Batu trachytes though not all of the Quaternary lavas are trachytic. On the plateau below Mt. Batu, highly vesicular, olivine-bearing Quaternary mafic lavas (see also Berhe *et al.*, 1987) blanket the Miocene sequence.

Bale Mountains mafic lavas are petrographically diverse. Miocene Bale basalts range from hypocrystalline to pyroxene megacryst-bearing lavas. All mafic units have olivine and clinopyroxene phenocrysts in a groundmass containing plagioclase, olivine and clinopyroxene, while interlayered trachytic units are dominated by plagioclase phenocrysts and microlites. Sample BM-14 is a 0·5 m wide dike cutting a highly altered tuff unit. Sample BM-15 contains olivine and plagioclase phenocrysts and clinopyroxene megacrysts up to 2 cm in length in an aphyric groundmass. These minerals vary from euhedral to anhedral and display partially resorbed grain boundaries. This olivine–clinopyroxene-bearing lava is overlain by a unit containing conspicuous euhedral clinopyroxene megacrysts up to 3–4 cm in length in a highly altered groundmass. Above this, BM-17 and BM-19 have experienced less overall alteration and carry primary rather than xenocrystic mineral assemblages. These samples have phenocrysts of olivine and clinopyroxene in a groundmass of plagioclase, Fe–Ti oxides and clinopyroxene. Alteration is observed mainly in the olivine phenocrysts and in the groundmass.

Pliocene and Quaternary Bale lavas are porphyritic to aphanitic, with increasing vesicle populations up-section. The lowest sampled units contain olivine and sparse clinopyroxene phenocrysts and the mineral assemblage is dominated increasingly by plagioclase feldspar and clinopyroxene towards the top of the volcanic pile; the youngest lavas are, however, characterized by abundant glomerophyric olivine phenocrysts and less abundant plagioclase feldspar. The groundmass consists primarily

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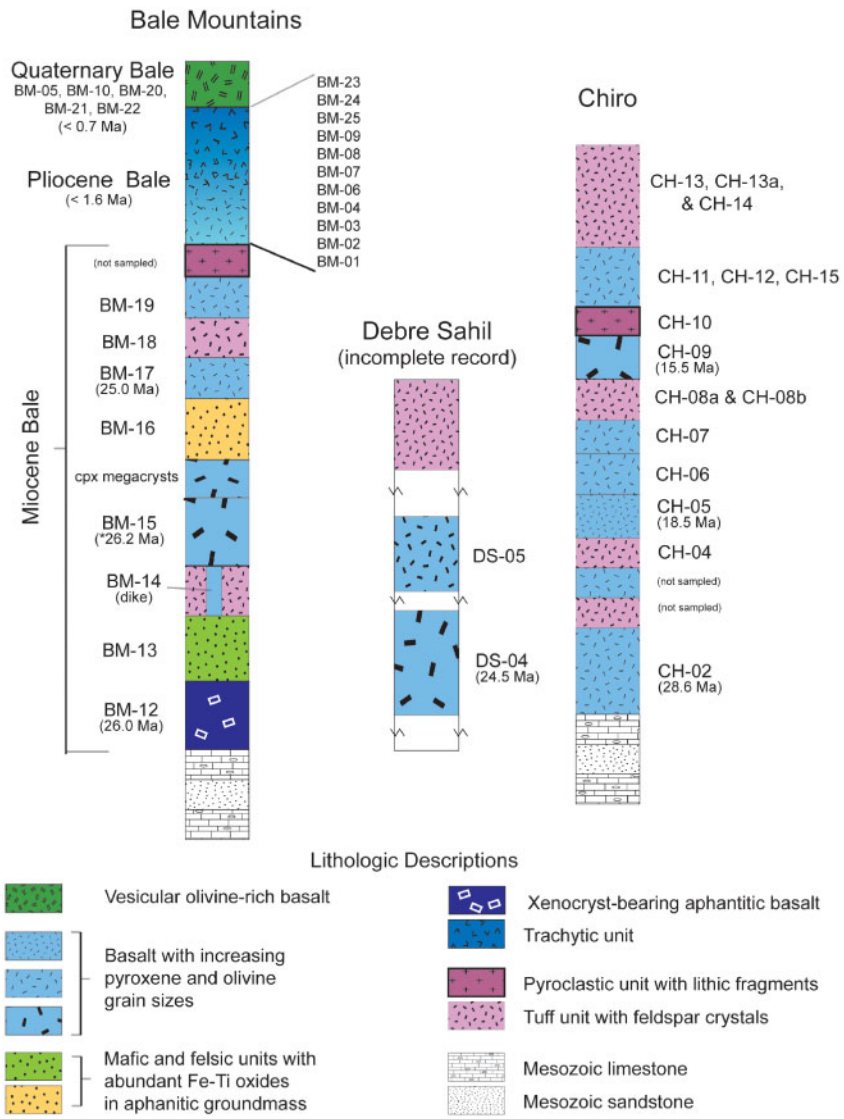


Fig. 3. Stratigraphic columns comparing the textures and petrography of the analysed samples. Sample numbers and $^{40}Ar/^{39}Ar$ ages are given where available. BM-14 is a small, highly altered dike cutting through the silicic tuff. Unit thicknesses are not to scale.

of plagioclase and clinopyroxene with less abundant Fe–Ti oxides and olivine.

METHODS

We focused on collecting mafic lavas; interlayered silicic units were collected but are not discussed here. Poor exposures, complex small-scale faulting and local subsidence made it difficult to determine the actual thickness of each unit. Sampling from the Bale Mountains and Chiro included the basal volcanic units overlying Mesozoic sedimentary units.

Each sample was cut into slabs to remove weathered surfaces, abraded to remove saw marks and then chipped using an alumina ceramic jaw crusher. Powders for major element, trace element and loss on ignition (LOI) analysis were generated in a tungsten carbide disc mill. All samples were prepared and analysed at Duke University based on methods outlined in Klein *et al.* (1991). Major and select trace element solutions were diluted to 1: 6250 and analysed using an ARL-Fisons Spectraspan 7 direct-coupled plasma mass spectrometer (DCP-MS). Analytical accuracy for replicate analysis of standards (USGS BHVO-1, BIR-1, W-2, QLO, G2, NBS/NIST 688 and LDEO AII-92) is <1% for SiO_2; <3% for Al_2O_3, Fe_2O_3*, MgO, CaO and Na_2O; <5% for TiO_2, MnO, K_2O, P_2O_5, Ni, V, Zr and Cr. All analyses have a precision < 3% RSD except for MnO, K_2O, P_2O_5 and Cu (<5% RSD); Cr, Zr, V and Zn (<10% RSD). Trace element concentrations were determined by analysing a 1: 30 000 sample dilution using a VG PlasmaQuad-3 ICP-MS. Standards with known concentrations and blanks were analysed after every 4 to 7 samples. Analytical accuracy was <5% RSD for all trace and rare earth elements except for Li, Cu, Rb, Cs, Eu, Dy, Ho, Lu, Pb, Th and U (<10% RSD).

High precision $^{40}Ar/^{39}Ar$ ages were determined for ten samples at Oregon State University. For each

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Table 1: $^{40}Ar/^{39}Ar$ age determinations

Sample	Plateau Age Spectrum					Total Fusion		Inverse Isochron Analyses		
	Age ±2σ (Ma)	^{39}Ar (%)	K/Ca	MSWD	Steps in plateau	Age ±2σ (Ma)	K/Ca	Age ±2σ (Ma)	$^{39}Ar/^{40}Ar$ Intercept	MSWD
CH-02	28.58 ± 0.33	75.1	0.062	0.32	8	29.93 ± 0.45	0.142	28.29 ± 0.57	311.2 ± 25	0.09
CH-05	18.51 ± 0.31	62.9	0.036	2.36	8	16.60 ± 0.30	0.096	19.69 ± 1.97	272.8 ± 39	0.85
CH-09	15.48 ± 0.48	72.7	0.090	2.78	5	14.89 ± 0.53	0.074	15.06 ± 0.78	301.0 ± 8.0	0.05
BM-03	0.87 ± 0.04	93.4	0.127	2.36	8	0.96 ± 0.05	0.320	0.89 ± 0.05	287.2 ± 28	0.26
BM-09	1.61 ± 0.03	88.1	0.305	2.57	6	1.73 ± 0.04	0.379	1.61 ± 0.05	293.2 ± 20	0.49
BM-12	26.01 ± 0.21	100.0	0.070	2.18	13	26.10 ± 0.28	0.297	25.96 ± 0.27	296.7 ± 3.9	1.12
BM-15	26.22 ± 0.48	47.6	0.016	2.45	7	30.79 ± 0.84	0.081	25.39 ± 1.02	323.1 ± 30	1.00
BM-17	25.02 ± 0.29	92.2	0.114	2.31	9	27.56 ± 0.79	0.268	24.63 ± 0.49	308.4 ± 13	0.40
BM-20	0.66 ± 0.11	93.2	0.063	2.36	8	0.82 ± 0.13	0.122	0.67 ± 0.43	294.5 ± 37	0.83
DS-04	24.45 ± 0.23	85.6	0.069	2.36	8	27.19 ± 0.40	0.199	23.86 ± 0.75	318.4 ± 28	0.53

sample, 2–3 g of 1–1·5 mm unweathered chips of basalt groundmass were hand-picked under a microscope, avoiding large olivine and pyroxene phenocrysts. See Koppers *et al.* (2011) for sample preparation techniques and analytical methodology details. All age plateaus are based on 47–100% of the ^{39}Ar released (Table 1; Supplementary Data Fig. S1; supplementary data are available for downloading at http://www.petrology.oxfordjournals.org). Additionally, all $^{40}Ar/^{36}Ar$ intercept values are within error of the atmospheric ratio (295·5 ± 0·7), suggesting excess ^{40}Ar is insignificant.

Helium isotopes were measured on five olivine-rich samples at Oregon State University. Olivine grains (~0·3 g) were hand-picked from crushed samples under a binocular microscope to avoid alteration and adhering groundmass. Analytical methodology is described in Graham *et al.* (2014).

Sr–Nd–Pb–Hf isotope determinations for 17 samples were performed at San Diego State University. Samples containing 4–14 wt % MgO were selected based on major and trace element chemistry. Emphasis was placed on higher MgO samples and those with Ce/Pb = 20–30, because lower ratios indicate likely crustal contamination (following Hofmann *et al.*, 1986; Hofmann, 1997). This criterion reduced the number of samples appropriate for isotopic analysis. For each sample, 350 mg of 1–2 mm phenocryst-poor chips were hand-picked under a binocular microscope. The chips were leached for 5 minutes using 2.5 N HCl, rinsed in ultra-pure H_2O, then leached using 2N HBr followed by an ultra-pure H_2O rinse. The chips were dissolved using a mixture of concentrated HNO_3 and concentrated HF over a period of 2–3 days. All of the isotopes were separated from the same batch of dissolved rock chips using ion exchange chromatography. Sr, Nd and Pb isotopes were separated for analysis following the procedures of Hanan & Schilling (1989) and Blichert-Toft *et al.* (1997) for Hf isotopes. Sr isotopes were measured on a VG Sector 54, seven-collector thermal ionization mass spectrometer in multi-dynamic mode. Hafnium, Nd and Pb isotopes were analysed on a Nu Plasma 1700 inductively coupled plasma mass spectrometry (MC-ICP-MS) system using a Nu Instruments desolvating nebulizer equipped with a 100 μl min^{-1} nebulizer (DSN100). The $^{87}Sr/^{86}Sr$ ratios were normalized to $^{88}Sr/^{86}Sr = 0·1194$ to correct for mass fractionation and the Sr isotopic data are reported relative to NIST SRM 987 = 0.710250. The Nd and Hf ratios were corrected for instrumental mass fractionation and machine bias by applying a discrimination factor determined by bracketing sample runs with analyses of the SDSU AMES Nd standard (every 2 samples) using $^{143}Nd/^{144}Nd = 0·512130$ and the JMC 475 Hf standard (every two samples) with $^{176}Hf/^{177}Hf = 0·282162$. Hf isotope sample data are normalized to $^{179}Hf/^{177}Hf = 0·7325$ and reported relative to JMC 475 = 0·282162. Sample data were normalized to $^{146}Nd/^{144}Nd = 0·7219$ and are reported relative to the laboratory value of $^{143}Nd/^{144}Nd = 0·512130$ for the SDSU AMES Nd standard. The measured value of La Jolla Nd at SDSU = 0·511844. Lead was separated on two anion exchange columns, a 500 μl followed by a 200 μl column to maximize the Pb separation for optimum ICP-MS analyses. The Pb yields were determined to be ~100%, therefore, the columns do not have a fractionation effect. The in-run mass discrimination for Pb was determined with NIST NBS SRM 997 Tl with $^{205}Tl/^{203}Tl = 2·3889$ and NIST NBS SRM 981 (Todt *et al.*, 1996) using the method of White *et al.* (2000). The Pb isotope ratios were corrected for machine bias and drift using the Tl-doping technique with sample/standard bracketing with analyses of the NIST SRM 981 (every two samples) corrected for fractionation (White *et al.*, 2000; Thirlwall, 2002; Albarède *et al.*, 2004). Blank corrections were not significant and, therefore, not applied to the data.

Sample preparation and Re–Os isotopic analyses were performed at the Department of Terrestrial Magnetism, Carnegie Institute for Science, on a subset of samples analysed for Sr–Nd–Pb–Hf isotopes. Because Os is a compatible element and is generally present in higher abundances in more primitive lavas, samples with >8 wt % MgO were targeted for Re–Os isotopic analysis. Rhenium–osmium isotopes were measured on six mafic lavas spanning the geographic and temporal range of samples. To provide additional context, three Pliocene mafic lavas from Turkana, N. Kenya (Furman *et al.*, 2004) and four Quaternary alkaline basalts from the central MER (Furman *et al.*, 2006a; Rooney *et al.*, 2007) were also

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analysed. Rock chips were generated with an alumina jaw crusher and then were finely powdered using an alumina shatterbox. For each sample, 1·4–1·5 g of powder was spiked with dilute ^{190}Os spike (0·4755 ng/g) and a dilute ^{185}Re spike (3 ng/g in 6 M HCl) and digested using the sealed Carius tube technique, described most recently by Class *et al.* (2009) and Heinonen *et al.* (2010). Osmium was extracted using carbon tetrachloride and purified by microdistillation, while Re was separated using ion exchange chromatography. The separated Os was loaded onto fused Pt ribbons and analysed using a Thermo-Fisher Triton TIMS operated in negative ionization mode. The separated Re was diluted using 0.5 M HNO_3 and analysed using a VG P-54 ICP-MS. Os isotope ratios were corrected for mass fractionation using $^{192}Os/^{188}Os = 3·082614$. To monitor accuracy and precision, the Johnson-Matthey (J-M) Os standard ($^{187}Os/^{188}Os = 0·1738$) was measured five times throughout the analyses and produced $^{187}Os/^{188}Os = 0·17372 \pm 0·00012$ (2SE). Osmium blanks ranged from 1–5 pg while Re blanks were consistently 5 pg.

RESULTS

Age determinations

$^{40}Ar/^{39}Ar$ age determinations (Table 1) record several periods of basaltic volcanism along the southeastern flank of the Ethiopian Plateau. These episodes post-date the primary flood basalt eruptions and they continued at some localities until the Quaternary. The basal Chiro lava CH-02 erupted near the end of Afar flood basalt volcanism (28·6 ± 0·33 Ma) and is overlain by younger Miocene basaltic flows (18·5 ± 0·31 and 15·5 ± 0·48 Ma). The youngest lavas in this area are Quaternary (Tefera *et al.*, 1996); due to their felsic compositions, they were not analysed for $^{40}Ar/^{39}Ar$ ages. Sample DS-04, from near the base of the Debre Sahil lava sequence, yielded an $^{40}Ar/^{39}Ar$ age of 24·5 ± 0·23 Ma. Bale Mountains lavas record two volcanic episodes: one is Miocene (26·01 ± 0·21 Ma to 25·02 ± 0·29 Ma) and the other is much younger (1·61 ± 0·03 to 0·66 ± 0·11 Ma). The younger lavas lie unconformably on the older lavas.

Major elements

Lavas from the southeastern Ethiopian Plateau range from primitive basanites and basalts (Table 2; Supplementary Data Fig. S2) to rhyolitic tuffs. Most Chiro transitional basalts have normative quartz and hypersthene; basal unit CH-02 is olivine-hypersthene normative, while olivine-rich CH-09 is olivine–diopside normative. Chiro lavas typically have <6 wt % MgO. They display coherent major element variations similar to those of HT1 Oligocene Afar plume lavas and appear to be dominated by low-pressure fractionation of the observed mineral phases (Fig. 4); sample CH-09 is a consistent outliner due to its high MgO content and visible, abundant olivine.

Miocene Debre Sahil and Bale Mountains lavas comprise a heterogeneous suite of mildly alkaline basalts, basanites, trachybasalts and basaltic trachyandesites (Fig. S2) and extend to more primitive compositions (6–10 wt % MgO) than are observed at Chiro. Lavas with <7 wt % MgO are more silica-undersaturated than Afar plume flood basalts (Supplementary Data Fig. S2). They are olivine–nepheline normative with notable exceptions: the two transitional lavas, BM-17 and BM-19, have little to no normative nepheline; the oldest units BM-12 and BM-13 contain normative leucite. Miocene Debre Sahil and Bale lavas lack internally coherent major element fractionation trends and instead show broad within-suite variations. Coherent trends in K_2O suggest crustal contamination or fractionation controlled K_2O abundances (Fig. 4). The Miocene Bale samples define two groups: most lavas are mildly alkaline with high CaO, MnO, Na_2O, CaO/Al_2O_3 and Zr (BM-12, BM-13, BM-14, BM-15, BM-16), but the two transitional lavas (BM-17 and BM-19) have lower abundances of these elements (Fig. 4). Pliocene and Quaternary Bale lavas define similar compositional trends, but their lower CaO/Al_2O_3 values suggest an earlier onset of clinopyroxene fractionation compared to alkaline Miocene Bale lavas and those at Chiro. Overall, compositional variations among the Pliocene and Quaternary Bale lavas are consistent with progressive removal of the mineral phases observed within each sample suite.

Trace elements

Compatible trace elements

Trace element variations in the three sample suites are related to distinct patterns of crystal fractionation (Fig. 5). Most Chiro lavas contain <100 ppm Cr and <60 ppm Ni, indicating removal of olivine after magma separation from the mantle (Table 2). At Debre Sahil, Ni and Cr concentrations in highly magnesian lava DS-04 are 310 ppm Ni and 932 ppm Cr, consistent with the abundant olivine phenocrysts observed in the sample. Other Debre Sahil and Bale lavas include both mildly fractionated (Ni <90 ppm and Cr ≤233 ppm) as well as silica-rich types.

Incompatible trace elements

Incompatible trace element abundances generally increase with decreasing MgO, although the magnitude of the increase varies for different elements and between sample suites. Miocene lavas in all suites encompass the full compositional range; alkaline Bale lavas have the highest abundances for most of the incompatible trace elements, while transitional basalts (samples BM-17, BM-19 plus most Chiro lavas) consistently have notably lower abundances of most elements for a given MgO content (Fig. 5). Comparing all suites, La vs MgO shows considerable scatter, suggesting that simple crystal fractionation is not the only process controlling

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Table 2: Major and trace elements for Southeast Ethiopian Plateau lavas

Locality	Chiro								
Sample	CH-02	CH-04	CH-05	CH-06	CH-07	CH-08a	CH-08b	CH-09	CH-10
Latitude (N)	9·171	9·152	9·057	9·018	8·817	8·843	8·843	8·845	8·911
Longitude (E)	41·060	41·023	40·888	40·900	40·502	40·456	40·456	40·458	40·482
SiO_2	46·03	56·54	49·54	50·24	49·84	70·38	70·25	46·35	65·89
TiO_2	3·92	1·55	3·27	3·21	3·31	0·62	0·70	1·91	1·08
Al_2O_3	14·00	17·57	14·31	14·54	14·33	12·80	14·52	14·09	14·21
Fe_2O_3*	14·97	8·30	14·15	13·36	13·75	4·45	4·14	12·23	5·79
MnO	0·21	0·12	0·21	0·22	0·20	0·20	0·13	0·15	0·22
MgO	5·73	1·76	5·35	4·77	4·76	0·39	0·35	10·38	0·93
CaO	9·78	4·58	10·53	8·61	9·35	0·33	0·49	12·04	1·32
Na_2O	2·98	5·31	2·65	3·23	3·22	5·55	6·42	1·98	5·30
K_2O	0·97	2·66	0·97	1·31	1·28	4·14	3·73	0·59	4·00
P_2O_5	0·71	0·71	0·42	0·57	0·52	0·06	0·10	0·18	0·22
Total	99·30	99·09	101·40	100·05	100·55	98·93	100·84	99·95	98·95
LOI	−0·15	−0·07	−0·10	−0·03	−0·10	−0·11	−0·17	−0·09	−0·11
Cs	0·12	0·14	0·26	0·69	0·25	0·18	0·18	0·03	1·13
Rb	12·0	44·5	19·7	29·7	16·1	114	69·9	7·00	75·4
Ba	513	869	221	417	325	288	745	115	1084
Th	1·81	8·42	2·85	3·91	3·54	16·6	12·1	1·02	11·6
U	0·54	1·85	0·68	0·85	0·74	1·67	1·76	0·30	2·88
Nb	20·2	75·0	27·5	25·8	32·1	138	126	12·2	96·8
Pb	2·01	8·59	3·11	5·36	3·57	15·3	9·48	1·23	11·8
Sr	724	752	512	516	479	5	50	367	91
Zr	158	340	270	279	233	943	905	109	580
Hf	3·91	7·70	6·07	6·25	5·63	22·3	20·0	2·78	15·8
Y	29·3	37·4	34·8	36·0	30·2	127	74·4	18·9	74·9
Cr	47·5	2·20	9·88	51·3	15·9	1·35	1·66	505	3·32
Ni	40·2	0·00	29·2	37·5	17·2	1·32	0·36	168	4·46
Sc	26·4	8·93	33·3	26·5	23·1	9·32	5·99	35·4	12·6
V	381	34·0	295	327	293	11·3	5·03	316	43·0
Co	54·5	14·4	64·5	51·3	46·4	7·65	12·3	72·6	10·1
Cu	54·1	10·5	106	72·1	71·0	3·37	3·81	91·6	30·9
Li	5·74	13·9		6·42	4·55			3·50	0·61
Be	1·03	3·02	1·52	1·67	1·61	5·65	5·32	0·61	0·01
Ta			1·75			8·38	7·38		
Mo			1·35			0·37	1·17		
Zn			137			167	185		
La	21·6	69·5	25·9	29·8	30·2	108	64·2	10·6	82·8
Ce	51·0	130	59·5	66·0	65·0	186	110	24·7	172
Pr	7·75	17·1	8·17	9·44	8·92	26·0	17·9	3·56	22·5
Nd	34·9	61·1	34·2	38·9	35·4	91·2	65·2	15·2	78·4
Sm	7·84	10·9	8·22	8·66	7·56	19·1	13·8	3·79	15·7
Eu	2·65	3·11	2·56	2·85	2·25	3·79	3·47	1·27	4·05
Gd	7·98	10·6	7·94	8·89	7·57	18·3	13·0	4·28	14·4
Tb	1·11	1·39	1·25	1·26	1·07	3·18	2·20	0·62	2·39
Dy	5·82	7·08	6·91	6·96	5·91	19·2	13·0	3·57	14·1
Ho	1·11	1·36	1·29	1·32	1·14	4·04	2·71	0·72	2·75
Er	2·69	3·53	3·22	3·40	2·83	11·2	7·37	1·71	7·37
Yb	2·14	3·06	2·67	2·87	2·36	11·2	7·54	1·51	7·31
Lu	0·34	0·45	0·38	0·44	0·35	1·71	1·16	0·23	1·10
normative minerals	ol-hy	qtz-hy	qtz-hy	qtz-hy	ol-hy	qtz-hy	qtz-hy	di-ol	qtz-hy

(continued)

magma evolution (Fig. 5). Chiro lavas have low La for their MgO content compared to other suites.

Abundances of Ba, Rb, Th, U and Pb are positively correlated with La among mafic lavas from all groups. However, those correlations vary in each lava suite (Fig. 5). Pliocene Bale lavas have low Th for their La abundance. Zirconium and Hf correlate positively with La, but Chiro lavas have high Zr abundances compared to most Debre Sahil and Bale lavas (Fig. 5). Niobium and Ta abundances show consistent positive correlations with La, though Quaternary Bale lavas are offset to lower La values for a given Nb abundance. Ethiopian Plateau lavas record a range of Ce/Pb ratios; all but the Debre Sahil lavas record values that are consistent with mantle-derived melts (25 ± 5; Hofmann *et al.*, 1986). However, none of the lava suites display Ce/Pb–MgO correlations (Fig. 6). Lavas with Ce/Pb values less than 20 are often ascribed to crustal contributions due to preferential addition of Pb over Ce. All but two Chiro lavas have Ce/Pb ratios less than 20, while none of the Debre Sahil lavas have mantle values.

The plateau lavas have Ba/Th–K/Nb and Ba/Nb–La/Nb relationships that suggest regional differences in magma sources (Fig. 6). Except for CH-02, Chiro lavas show a

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Table 2: Continued

							Debre Sahil		
Sample	CH-11	CH-12	CH-13a	CH-13b	CH-14	CH-15	DS-01	DS-02	DS-03
Latitude (N)	8·922	8·925	9·167	9·167	9·149	9·116	8·389	8·389	8·389
Longitude (E)	40·422	40·401	40·801	40·801	40·801	40·850	39·942	39·942	39·942
SiO_2	51·81	50·81	71·34	71·40		49·44	46·92	50·99	48·30
TiO_2	2·75	2·63	0·36	0·39		3·13	1·94	1·76	1·67
Al_2O_3	13·62	13·34	15·82	14·60		14·30	15·72	18·25	14·72
Fe_2O_3*	13·04	14·08	2·62	2·72		14·33	13·86	9·91	12·31
MnO	0·21	0·20	0·06	0·02		0·22	0·25	0·18	0·19
MgO	4·61	4·73	0·53	0·19		4·37	5·07	3·58	7·50
CaO	9·01	8·65	0·92	0·63		8·80	9·30	7·74	9·01
Na_2O	2·86	2·83	2·24	5·26		3·50	3·81	4·69	3·26
K_2O	1·65	1·55	4·58	4·14		1·63	2·07	2·68	2·05
P_2O_5	0·37	0·36	0·07	0·06		0·42	0·63	0·80	0·61
Total	99·92	99·18	98·54	99·41		100·13	99·57	100·58	99·63
LOI	−0·10	−0·14	−0·05	−0·10	−0·11	0·00	0·02	0·02	0·02
Cs	0·22	0·24	1·12	0·93	0·42	0·36	0·62	0·27	0·30
Rb	25·03	23·97	112·46	110·35	103·72	29·16	59·39	38·44	24·49
Ba	373	354	401	488	42	356	1699	978	882
Th	3·01	3·02	15·92	8·12	16·22	3·18	4·65	3·30	1·72
U	0·72	0·70	3·48	1·64	2·13	0·82	0·85	0·67	0·37
Nb	25·6	25·1	91·3	89·7	87·9	29·8	49·9	37·2	20·2
Pb	4·35	4·77	12·13	11·13	11·29	4·18	11·24	7·08	5·43
Sr	361	352	102	112	3	507	1539	970	860
Zr	270	261	571	462		265	173	234	161
Hf	6·44	6·56	13·35	6·89	20·26	6·37	7·46	5·30	3·79
Y	41·44	40·37	54·50	48·13	89·38	37·11	50·87	30·75	24·16
Cr	70·83	72·30	2·35	4·17	2·10	2·03	17·20	2·11	159·38
Ni	42·67	46·44	2·18	2·13	1·63	7·24	42·66	4·60	60·45
Sc	31·81	31·73	2·22	2·38	8·40	29·99	41·94	10·31	24·01
V	350·20	352·20	10·32	15·69	5·22	347·62	321·94	161·87	229·40
Co	50·19	53·45	10·83	48·70	17·87	76·07	83·20	33·19	49·27
Cu	145·76	149·85	3·30	3·63	3·77	44·68	344·23	32·66	68·50
Li	0·27	1·04			0·01	0·64	0·08	0·04	0·02
Be	3·96	14·63	5·75	4·79	0·05	1·74	0·06	0·00	0·04
Ta			6·12	5·71					
Mo			4·94	2·52					
Zn			105	72					
La	28·6	28·3	77·9	83·8	85·4	30·3	55·4	41·0	29·6
Ce	62·5	61·7	141·6	140·8	181·9	67·0	116·6	83·6	62·9
Pr	8·80	8·84	17·69	19·66	23·99	9·37	15·16	10·44	8·21
Nd	36·42	35·46	60·13	66·87	81·30	38·37	56·65	37·74	30·72
Sm	8·41	8·02	11·82	12·93	17·00	8·51	11·60	7·48	6·11
Eu	2·35	2·34	1·63	1·96	1·44	2·51	4·08	2·47	2·09
Gd	8·91	8·70	10·38	10·74	14·90	8·75	10·92	6·70	5·49
Tb	1·31	1·30	1·76	1·73	2·62	1·25	1·71	1·02	0·85
Dy	7·58	7·50	9·92	9·05	16·02	7·00	9·77	5·75	4·55
Ho	1·51	1·50	1·97	1·67	3·33	1·37	1·90	1·14	0·89
Er	3·97	3·89	5·32	4·34	9·09	3·40	4·75	2·92	2·34
Yb	3·56	3·46	5·36	3·80	9·63	3·04	4·59	2·72	2·17
Lu	0·55	0·54	0·79	0·54	1·45	0·47	0·69	0·41	0·34
normative minerals	qtz-hy	qtz-hy	qtz-hy (cor?)	qtz-hy (cor?)	N/A	ol-hy	ne-ol-di	ne-ol-di	ne-ol-di

(continued)

tight correlation in Ba/Th–K/Nb space, distinct from other lava suites. Miocene Bale lavas show no correlation between Ba/Th–K/Nb, while Debre Sahil, Pliocene Bale and Quaternary Bale lavas show variable positive correlations. Pliocene Bale lavas record distinctly higher Ba/Th values, consistent with their low Th contents. Three Debre Sahil lavas, one Quaternary Bale lava (BM-23) and one Chiro lava (CH-02) also have elevated Ba/Th values, with some also trending to high K/Nb values.

Chiro lavas also have a shallow-sloped Ba/Nb–La/Nb correlation that overlaps those of Oligocene HT1 flood basalts (Pik *et al.*, 1999). Alkaline Miocene Bale lavas, Debre Sahil, Pliocene Bale and Quaternary Bale lavas define a steeper Ba/Nb–La/Nb trend than Chiro lavas and overlap the field of LT basalts. Transitional Miocene Bale lavas (BM-17 and BM-19) have low Ba/Nb values, similar to those for 26–16 Ma Turkana lavas (Furman *et al.*, 2006b) and HT2 lavas (Pik *et al.*, 1998). In contrast, Quaternary Bale lavas have low La/Nb (<0·6) and a

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Table 2: Continued

			Miocene Bale Mountains					
Sample	DS-04	DS-05	BM-12	BM-13	BM-14	BM-15	BM-16	BM-17
Latitude	8·363	8·352	6·882	6·882	6·884	6·885	6·887	6·890
Longitude	39·874	39·876	40·293	40·299	40·298	40·299	40·302	40·297
SiO_2	44·42	43·89	41·95	40·58	47·55	44·02	45·27	47·30
TiO_2	2·06	2·77	2·91	2·26	2·93	2·57	2·17	2·46
Al_2O_3	9·84	12·78	13·58	11·19	17·03	12·36	15·23	13·14
Fe_2O_3*	13·81	13·15	15·95	15·14	12·33	14·18	14·23	13·77
MnO	0·21	0·20	0·28	0·26	0·16	0·22	0·29	0·16
MgO	14·33	7·97	6·14	9·11	4·24	9·32	4·87	9·88
CaO	11·26	12·01	11·55	14·57	8·29	13·73	10·13	8·88
Na_2O	2·67	4·32	5·61	4·07	3·02	3·45	4·90	2·82
K_2O	1·18	1·64	2·02	2·24	3·00	0·74	2·27	1·36
P_2O_5	0·53	0·71	0·82	0·85	1·00	0·55	0·99	0·40
Total	100·31	99·46	100·82	100·25	99·53	101·13	100·34	100·17
LOI	0·03	0·01	−0·11	0·00	0·01	0·01	0·03	0·03
Cs	0·65	0·41	0·64	0·38	0·62	0·54	0·42	0·30
Rb	9·08	39·62	26·95	47·58	57·79	12·88	45·72	19·62
Ba	469	614	875	957	1467	664	814	238
Th	2·83	5·45	10·11	8·31	9·20	4·88	5·57	3·10
U	0·62	1·35	2·05	1·68	2·20	1·02	1·10	0·84
Nb	34·1	62·9	92·9	54·6	116·8	52·2	52·4	34·5
Pb	4·00	5·20	5·27	6·17	6·16	5·95	9·09	2·48
Sr	485	646	947	924	1317	897	1536	481
Zr	145	230	289	148	378	251	185	255
Hf	3·66	5·70	6·55	3·53	7·88	4·58	4·40	5·34
Y	19·16	27·74	33·12	27·00	36·14	24·72	34·69	22·36
Cr	931·70	232·87	17·13	209·40	2·22	333·18	3·85	327·98
Ni	310·11	89·52	17·73	100·04	6·03	88·98	5·23	260·85
Sc	32·05	30·14	13·78	24·72	10·39	28·84	11·55	21·32
V	259·66	323·06	299·11	321·32	180·98	310·64	222·66	205·07
Co	76·01	99·99	55·43	64·22	41·14	62·91	44·00	74·99
Cu	166·64	178·38	94·18	142·26	28·68	104·73	69·31	77·22
Li	0·21	0·12	14·53	10·42	14·25		15·93	
Be	0·16	0·04	4·29	4·11	2·87	2·62	4·69	1·48
Ta						3·25		2·24
Mo						1·02		1·19
Zn						104		136
La	25·5	45·8	73·2	47·2	83·5	41·2	42·6	28·1
Ce	52·6	93·3	135·4	82·7	159·9	81·9	79·5	61·4
Pr	6·70	11·57	16·53	9·62	20·22	10·04	9·87	8·02
Nd	24·96	41·63	57·75	33·66	72·33	37·52	36·25	32·34
Sm	5·25	8·28	10·27	6·18	12·49	7·55	7·20	7·35
Eu	1·70	2·65	2·97	1·96	3·74	2·34	2·29	2·31
Gd	4·80	7·24	10·52	7·06	12·52	6·52	8·00	6·49
Tb	0·73	1·08	1·29	0·93	1·45	0·98	1·14	0·96
Dy	3·94	5·65	6·59	5·03	7·06	5·02	6·22	4·79
Ho	0·71	1·05	1·23	1·00	1·33	0·93	1·29	0·84
Er	1·78	2·57	3·15	2·54	3·41	2·30	3·42	1·95
Yb	1·47	2·17	2·60	2·22	2·75	1·93	3·09	1·47
Lu	0·21	0·31	0·40	0·35	0·42	0·29	0·49	0·21
normative minerals	ne-ol-di	ne-ol-di	ne-leu-ol-di	ne-leu-ol-di	ne-ol-di	ne-ol-di	ne-ol-di	ne-ol-di

(continued)

small range in Ba/Nb (Fig. 6). Outlier samples (three Debre Sahil lavas, one Quaternary Bale lava and CH-02) have notably elevated Ba/Nb and La/Nb values.

Chondrite-normalized rare earth element patterns have moderate slopes, with varying degrees of light/middle and middle/heavy REE enrichment (Fig. 7). With the exception of CH-04, Chiro lavas have a relatively shallow slope (La/Sm_N = 1·7–2·6; La/Yb_N = 5·0–9·2). Transitional Bale and Quaternary Bale have similar light/middle REE ratios, but trend towards higher light/heavy REE ratios (La/Yb_N up to 13·7). Debre Sahil and Pliocene Bale lavas have steeper slopes (La/Sm_N = 2·8–3·5; La/Yb_N = 8·7–15·3). Alkaline Bale lavas record the largest slope variations (La/Sm_N up to 4·9; La/Yb_N up to 21·8).

Radiogenic isotopes

He isotopes

Helium concentrations in olivine from five of the plateau lavas range from 1·5 to 4·5 *n*cm^3STP/g (Table 3), comparable to other EARS lavas (Marty *et al.*, 1996; Scarsi & Craig, 1996; Hilton *et al.*, 2011; Rooney *et al.*, 2012). The helium isotopic signatures, however, are

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Table 2: Continued

			Pliocene Bale Mountains						
Sample	BM-18	BM-19	BM-01	BM-02	BM-03	BM-04	BM-06	BM-07	BM-08
Latitude	6·890	6·891	7·282	7·282	7·282	7·282	7·007	6·980	6·933
Longitude	40·297	40·298	39·822	39·822	39·822	39·822	39·968	39·967	39·946
SiO_2	69·64	47·26	50·37	50·42	45·33	44·62	43·58	45·81	49·03
TiO_2	0·75	2·25	2·11	2·10	3·05	2·92	2·97	2·92	2·03
Al_2O_3	13·65	13·49	16·43	16·60	15·22	15·04	14·86	15·66	17·58
Fe_2O_3*	3·65	13·24	12·31	12·63	14·81	14·60	14·42	14·01	11·03
MnO	0·12	0·16	0·18	0·15	0·20	0·19	0·20	0·20	0·17
MgO	0·57	9·26	4·22	3·23	7·09	7·20	7·33	6·88	5·60
CaO	0·43	9·22	8·80	7·74	9·51	9·16	9·87	8·94	8·64
Na_2O	5·21	2·81	3·74	3·84	3·64	3·75	4·13	3·37	3·94
K_2O	4·35	1·03	2·22	2·12	1·60	1·72	1·70	1·43	1·70
P_2O_5	0·11	0·34	0·41	0·42	0·74	0·77	1·11	0·72	0·50
Total	98·48	99·06	100·79	99·26	101·17	99·98	100·18	99·94	100·24
LOI	0·02	0·02	0·01	−0·09	0·01	0·00	0·00	0·02	0·00
Cs	0·30	0·33	0·12	0·28	0·12	0·20	0·25	0·06	0·15
Rb	116·67	17·50	43·86	44·64	21·22	21·94	22·46	13·94	17·23
Ba	455	210	586	601	624	619	815	599	539
Th	15·68	2·73	5·19	4·93	2·39	2·59	2·87	1·67	2·13
U	1·51	0·73	1·11	0·67	0·52	0·61	0·64	0·25	0·50
Nb	99·8	29·3	46·2	43·2	37·3	39·6	58·8	29·1	49·6
Pb	17·60	2·28	5·24	5·55	3·07	3·44	3·69	3·11	3·74
Sr	19	456	509	545	1115	1092	1331	914	716
Zr	460	191	236	227	172	171	187	154	179
Hf	13·30	4·77	5·63	5·39	3·89	4·06	4·19	3·68	3·99
Y	41·76	23·58	29·26	30·84	26·72	28·26	29·89	31·54	25·19
Cr	5·23	293·51	15·27	21·21	91·41	102·55	119·20	110·24	16·86
Ni	7·94	238·00	19·80	29·00	66·78	76·24	85·48	82·26	30·42
Sc	6·51	22·83	21·40	21·78	19·83	21·41	19·87	24·88	20·19
V	11·04	209·80	241·88	237·87	240·98	251·90	238·11	264·33	203·83
Co	17·53	76·20	98·07	48·02	57·88	68·43	61·87	61·26	46·10
Cu	7·89	70·04	59·10	84·86	52·65	59·28	57·77	64·87	35·40
Li			3·41	0·09	3·14	5·00	4·12	2·30	1·56
Be	4·79	1·30	9·87	0·03	1·39	3·57	3·09	15·50	1·13
Ta	5·99	1·92							
Mo	0·66	1·13							
Zn	141	124							
La	94·4	23·9	41·3	40·4	32·3	33·8	49·0	29·0	29·0
Ce	192·9	52·3	81·2	81·2	67·9	70·9	99·9	60·5	58·7
Pr	22·54	6·91	10·32	10·13	9·23	9·74	13·20	8·38	7·71
Nd	78·34	28·11	38·30	36·42	36·61	38·93	50·49	34·41	29·37
Sm	14·55	6·55	7·35	7·22	7·42	7·65	9·00	7·03	5·78
Eu	3·30	2·11	2·11	2·21	2·47	2·58	3·02	2·45	2·06
Gd	11·31	6·15	7·69	6·65	7·56	7·93	9·61	7·75	6·20
Tb	1·76	0·95	1·02	1·01	1·00	1·04	1·18	1·05	0·86
Dy	8·82	4·92	5·59	5·59	5·26	5·54	6·02	5·71	4·63
Ho	1·61	0·88	1·08	1·08	0·99	1·05	1·12	1·14	0·91
Er	4·05	2·14	2·81	2·71	2·48	2·62	2·83	2·89	2·37
Yb	3·62	1·72	2·60	2·52	2·03	2·16	2·29	2·50	2·09
Lu	0·55	0·24	0·39	0·38	0·30	0·34	0·35	0·39	0·34
normative minerals	qtz-hy	ol-hy-di	ne-ol-di	ol-hy	ne-ol-di	ne-ol-di	ne-ol-di	ne-ol-di	ne-ol-di

(continued)

widely variable (Table 3). The olivine-rich Chiro lava CH-09 has $^3He/^4He = 9{\cdot}1\,R_A$, within the range for mid-ocean ridge basalts ($8{\cdot}1 \pm 1\,R_A$; Graham, 2002) and at the lower limit previously inferred for Afar plume lavas (9–19·5 R_A; Marty *et al.*, 1996; Scarsi & Craig, 1996; Pik *et al.*, 2006). Miocene Debre Sahil and Bale lavas have lower $^3He/^4He$ values (6·15 and 5·15 R_A, respectively), similar to mafic lavas from Turkana and Huri Hills in northern Kenya (5·9–7·8 R_A; Class *et al.*, 2005; Furman *et al.*, 2006b; Pik *et al.*, 2006; Locke *et al.*, 2008). Notably, two Quaternary Bale lavas have higher $^3He/^4He$ values (12·1 and 12·5 R_A) similar to plume-influenced Oligocene flood basalts near Afar, MER basalts and some Rungwe basalts (Marty *et al.*, 1996; Scarsi & Craig, 1996; Hilton *et al.*, 2011; Rooney *et al.*, 2012).

Sr, Nd and Pb isotopes

Sr, Nd and Pb isotopes show significant spatial and temporal variability. The northernmost Chiro lavas (28–15 Ma) have the highest values of both $^{87}Sr/^{86}Sr_i$ and $^{143}Nd/^{144}Nd_i$ (up to 0·7044, ~0·5129; Table 4, Fig. 8) and are isotopically similar to HT1 lavas (Pik *et al.*, 1999). Debre Sahil lavas have $^{87}Sr/^{86}Sr_i = 0{\cdot}7039$–$0{\cdot}7041$;

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Table 2: Continued

					Quaternary Bale Mountains				
Sample	BM-09	BM-23	BM-24	BM-25	BM-05	BM-10	BM-20	BM-21	BM-22
Latitude	6·927	6·882	6·898	6·924	7·147	7·084	7·026	7·038	7·069
Longitude	39·940	39·905	39·903	39·935	40·000	40·111	40·322	40·303	40·204
SiO_2	46·76	52·89	48·04	48·03	47·84	47·17	45·61	45·62	46·01
TiO_2	3·04	1·96	2·73	2·71	1·43	1·77	1·76	1·75	2·01
Al_2O_3	16·48	17·33	16·86	16·98	15·28	14·99	14·61	14·57	15·08
Fe_2O_3*	12·70	9·35	12·39	12·30	11·61	12·13	12·30	12·31	12·40
MnO	0·20	0·17	0·20	0·19	0·17	0·18	0·17	0·18	0·18
MgO	4·81	3·77	4·37	4·37	9·16	8·07	9·91	9·43	7·39
CaO	7·01	6·10	6·81	6·74	11·30	11·95	11·85	11·85	12·34
Na_2O	4·75	4·86	5·28	4·98	2·79	3·07	3·14	3·09	3·41
K_2O	1·89	2·94	2·08	1·96	0·74	1·03	1·02	1·08	1·33
P_2O_5	0·92	0·50	1·14	1·11	0·20	0·31	0·33	0·33	0·39
Total	98·55	99·86	99·91	99·38	100·51	100·66	100·69	100·21	100·54
LOI	0·00	−0·12	−0·17	−0·12	0·00	0·00	0·00	0·00	0·00
Cs	0·16	0·11	0·26	0·19	0·04	0·16	0·12	0·11	0·13
Rb	23·47	25·67	26·74	24·60	9·46	16·00	16·53	14·97	19·06
Ba	747	1221	866	865	294	342	309	311	369
Th	3·17	3·10	3·08	3·03	1·06	2·33	1·90	1·91	2·49
U	0·70	0·75	0·73	0·69	0·17	0·61	0·38	0·38	0·52
Nb	55·1	39·5	59·3	58·3	18·7	38·2	38·4	40·9	51·8
Pb	3·45	5·28	3·54	3·50	1·64	2·60	1·91	1·84	2·19
Sr	1370	618	1388	1423	358	479	492	472	531
Zr	197	233	202	198	89	133	137	138	163
Hf	4·54	5·25	4·28	4·43	2·29	3·22	3·39	3·31	3·79
Y	34·07	33·11	32·49	36·37	18·72	22·87	21·29	21·39	24·11
Cr	2·51	16·54	2·07	1·89	307·42	228·42	334·01	309·27	176·95
Ni	5·58	20·82	3·94	3·51	139·26	113·93	169·64	156·64	88·98
Sc	14·68	17·88	12·40	12·57	26·95	27·67	28·65	27·70	27·19
V	150·09	167·25	119·14	122·08	230·99	237·14	241·29	236·30	258·17
Co	32·72	28·48	38·24	31·68	80·71	59·03	64·29	62·85	58·90
Cu	35·78	28·35	31·99	35·28	83·57	93·71	100·44	97·85	109·61
Li	0·05	0·17	0·02	0·02	1·41	19·13	6·68	5·96	6·14
Be	0·00	0·05	0·01	0·02	9·16	1·09	1·05	1·00	1·23
Ta									
Mo									
Zn									
La	41·4	41·2	46·3	50·3	11·0	20·7	19·5	19·6	24·2
Ce	87·8	83·9	96·3	97·8	23·5	41·9	41·2	41·0	49·7
Pr	11·40	10·78	12·32	12·72	3·19	5·51	5·47	5·46	6·59
Nd	43·23	40·08	45·41	46·68	13·61	21·71	21·88	21·86	26·26
Sm	8·77	7·85	8·82	8·96	3·12	4·61	4·76	4·80	5·54
Eu	2·94	3·14	3·03	3·09	1·13	1·56	1·51	1·49	1·71
Gd	8·03	7·46	7·69	8·08	3·78	5·20	4·98	4·85	5·93
Tb	1·20	1·14	1·17	1·22	0·57	0·73	0·71	0·69	0·83
Dy	6·51	6·20	6·11	6·46	3·43	4·26	4·11	4·10	4·55
Ho	1·26	1·22	1·15	1·24	0·69	0·83	0·78	0·77	0·90
Er	3·13	3·08	2·92	3·17	1·75	2·16	2·00	1·95	2·27
Yb	2·87	2·88	2·70	2·84	1·63	1·91	1·72	1·72	2·02
Lu	0·42	0·44	0·41	0·44	0·26	0·31	0·27	0·26	0·32
normative minerals	ne-ol-di	ne-ol-di	ne-ol-di	ne-ol-di	ne-ol-di	ne-ol-di	ne-ol-di	ne-ol-di	ne-ol-di

(continued)

$^{143}Nd/^{144}Nd_i$ = 0·51266–0·51276 and Miocene Bale lavas have $^{87}Sr/^{86}Sr_i$ = 0·7033–0·7036; $^{143}Nd/^{144}Nd_i$ = 0·51271–0·51282; both groups plot within the fields of southern Ethiopian mafic lavas (George & Rogers, 2002). Transitional Miocene Bale lavas have slightly higher $^{143}Nd/^{144}Nd_i$ (0·51278–0·51280) values than their alkaline counterparts, but fall within the same range of $^{87}Sr/^{86}Sr_i$ values (0·7034–0·7035). Pliocene and Quaternary Bale lavas have more tightly correlated isotopic compositions; Pliocene lavas plot within the field of young MER basalts, while Quaternary Bale lavas have compositions that overlap with the proposed composition of the Afar plume (Rooney *et al.*, 2012) (Fig. 8).

The Pb isotopic data display clear temporal and spatial patterns. Chiro isotope signatures are scattered above and adjacent to the Northern Hemisphere Reference Line (NHRL). Their Pb isotopic values ($^{206}Pb/^{204}Pb_i$ = 18·5–18·9; $^{207}Pb/^{204}Pb_i$ = 15·55–15·57; $^{208}Pb/^{204}Pb_i$ = 38·5–38·9) are less radiogenic than the proposed end-member composition of the Afar plume (Rooney *et al.*, 2012) and trend towards the Pan-African lithosphere (Fig. 8). Debre Sahil and Miocene Bale lavas are more radiogenic and form an array that is sub-parallel to and slightly above the NHRL, approaching compositions of HIMU-like Miocene Turkana lavas and other post-flood basalt mafic lavas from Bure, the

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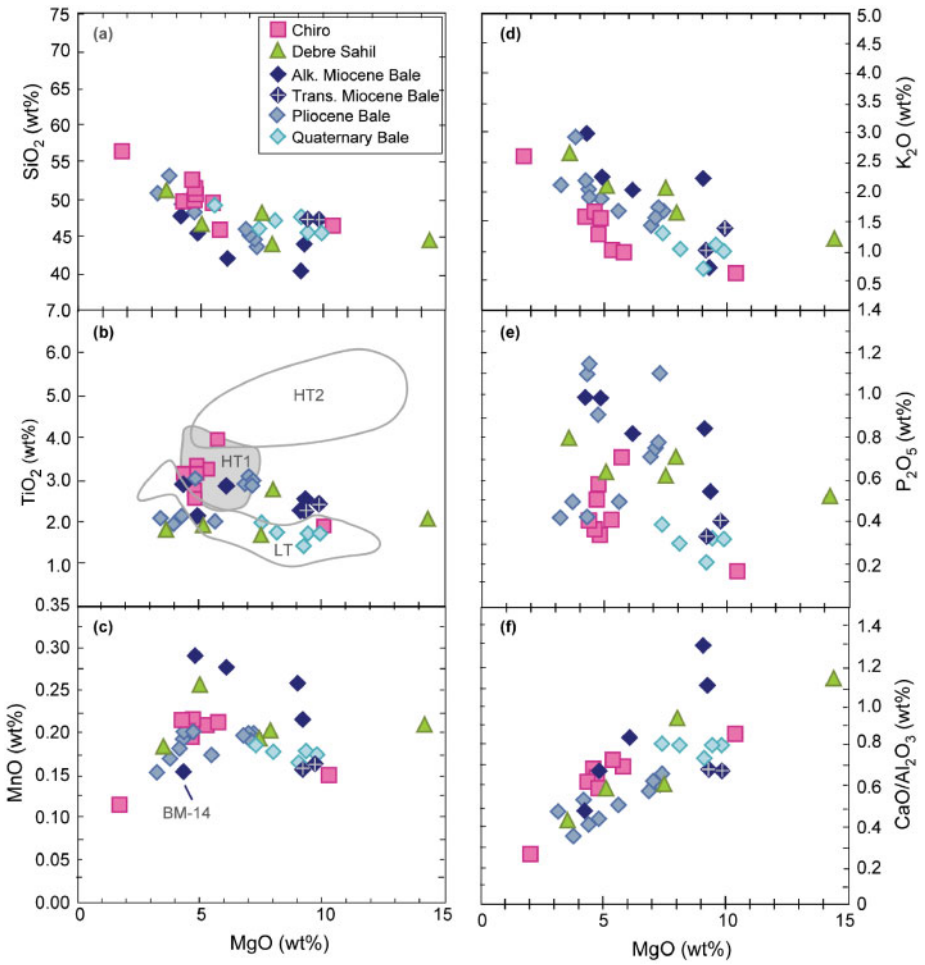


Fig. 4. Major element variation diagrams. Alkaline and transitional Miocene Bale lavas are separated for clarity. Fields for HT1, HT2 and LT lavas are from Pik *et al.* (1998, 1999) and Beccaluva *et al.* (2009).

Choke and Guguftu shield volcanoes (NW Ethiopia) and the Turkana depression in southern Ethiopia and northern Kenya. Transitional basalts BM-17 and BM-19 have lower radiogenic Pb isotopic signatures than alkaline Bale lavas and are isotopically similar to Afar plume lavas. Debre Sahil lavas have mildly elevated $^{208}Pb/^{204}Pb$ ratios, similar to 26 Ma Gerba Guracha lavas that were derived by melting of a source containing hydrous metasomes (Rooney *et al.*, 2014). Pliocene and Quaternary Bale lavas have Pb isotopic signatures overlapping the field of MER basalts.

Hf isotopes

All southeastern Ethiopian Plateau lavas lie near the ε_{Hf}–ε_{Nd} mantle array, but each sample suite is distinct (Table 4; Fig. 9). Chiro lavas have compositions similar to the proposed Afar plume composition (Rooney *et al.*, 2012) and trend toward coeval Bure basalts and lavas from Saudi Arabia that are interpreted as mixtures of the Afar plume and the subcontinental lithospheric mantle (Bertrand *et al.*, 2003). Debre Sahil and alkaline Miocene Bale lavas plot below the mantle array, with ε_{Nd}–ε_{Hf} values like Turkana picrites of similar age (Locke *et al.*, 2008) and they extend toward unradiogenic Hf and Nd isotopic ratios. Transitional Bale lavas BM-17 and BM-19 plot just below the mantle array (ε_{Hf} = 6·1–6·8, ε_{Nd} = 3·2–3·5). Alkaline Miocene Bale and Debre Sahil lavas record strong negative correlations between Nd and Hf isotopes with $^{206}Pb/^{204}Pb_i$ (Fig. 9). The Chiro, Miocene Bale and Debre Sahil lavas are also distinct from dikes of similar age (31–22 Ma) that were emplaced along the northwestern MER margin (Rooney *et al.*, 2013); Pliocene and Miocene Bale lavas overlap the ε_{Hf} – $^{206}Pb/^{204}Pb_i$ field for the 31–22 Ma dikes (Fig. 9). The young (< 10 Ma) MER dikes, however, have higher ε_{Hf} and ε_{Nd} than young Bale lavas.

Re–Os isotopes

With one exception, all the lavas from this study have low Os concentrations (5–51 ppt; Table 5). Their age-corrected $^{187}Os/^{188}Os$ values are more radiogenic (> 0·148) than those of the 30 Ma Ethiopian HT2 flood basalts (0·1239–0·1329; Rogers *et al.*, 2010; Nelson *et al.*, 2012) and of Miocene Turkana lavas (0·1450–0·1483; Nelson *et al.*, 2012) (Fig. 10). The olivine-rich Debre Sahil lava (DS-04) has the highest Os

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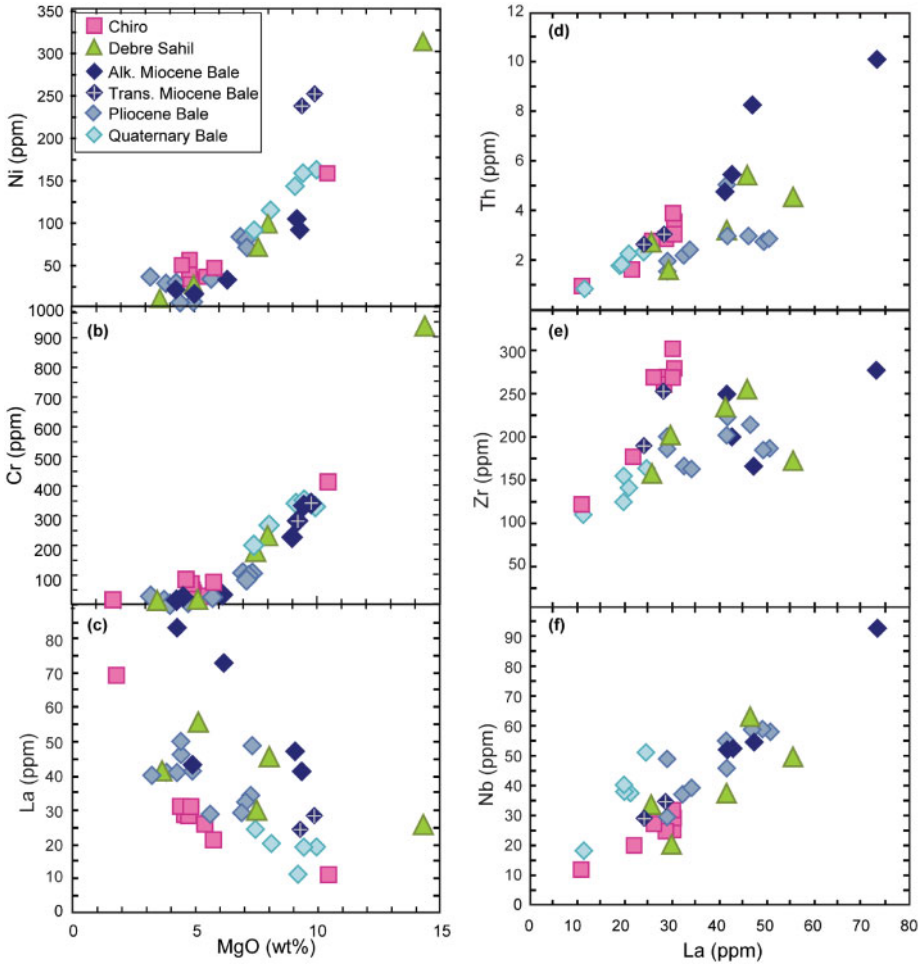


Fig. 5. Select compatible and incompatible trace element variation diagrams. The Chiro and Debre Sahil lavas with elevated wt % MgO have accumulated olivine.

concentration (195 ppt), a moderate Re abundance (281 ppt) and $^{187}Os/^{188}Os_i = 0{\cdot}1757$. The basal Miocene Bale unit (BM-12) has extremely low Os (5 ppt), while the other two Miocene Bale lavas have 18 and 31 ppt Os and $^{187}Os/^{188}Os_i = 0{\cdot}1484$ and 0·1754, respectively (Table 5). Quaternary Bale lava BM-20 has low Os (12 ppt), low Re (44 ppt) and the highest $^{187}Os/^{188}Os_i$ value (0·2266). While this Os isotope ratio is highly radiogenic, it is within the range of $^{187}Os/^{188}Os_i$ observed in young MER and Turkana lavas. There are no obvious correlations between $^{187}Os/^{188}Os_i$ and MgO, Ni, Os, Re, age or sample location. Miocene Bale lavas do show a positive correlation between $^{187}Os/^{188}Os_i$, $^{206}Pb/^{204}Pb_i$ and Ba/Rb and a distinct negative correlation with $^{143}Nd/^{144}Nd_i$ and K/Nb (Fig. 10).

DISCUSSION

Petrogenesis of southeastern Ethiopia Plateau basalts

In order to probe the origin and distribution of the various source domains that contribute to Ethiopian mafic volcanism, we first explore the effects of shallow-level differentiation and crustal contamination. These insights help identify the thermal conditions under which melting and magma emplacement have occurred and enable further examination of the depth and degree of mantle melting, as well as the nature and composition of the source materials.

Crystal fractionation

Each volcanic group records a distinct evolutionary history reflected by the composition of their parental melts and the onset of cotectic crystallization of olivine, pyroxene, Fe–Ti oxides and plagioclase. As evidenced by moderate Ni (17–260 ppm) and MgO (<10·5 wt %) contents, there are no primary mantle melts observed among the southeastern Ethiopian Plateau basalts; two samples (CH-09 and DS-04) have accumulated olivine. Linear correlations in major element abundances in Chiro and younger Bale lavas demonstrate the importance of fractional crystallization in those suites (e.g. SiO_2 and K_2O vs MgO, Fig. 4). In contrast, the lack of crystal fractionation trends between and within the mildly alkaline and transitional Miocene Bale lava groups preclude a simple genetic relationship (e.g.

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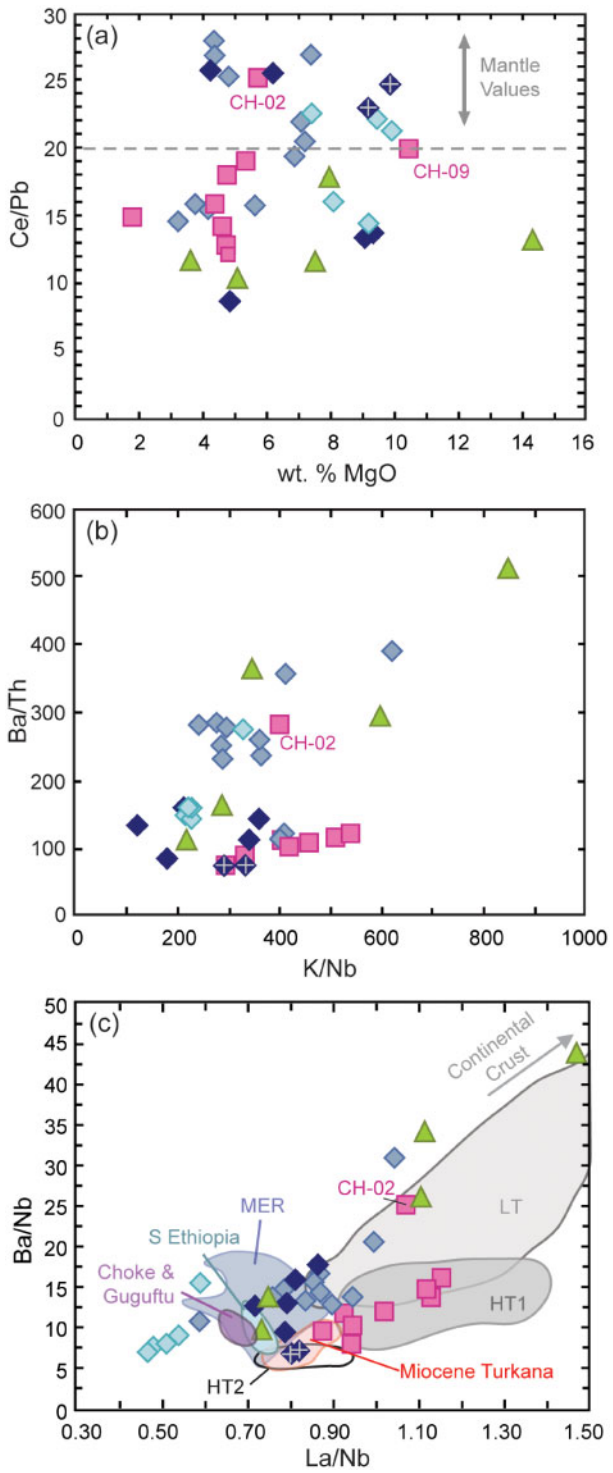


Fig. 6. (a) Ce/Pb vs wt % MgO for southeastern Ethiopian Plateau lavas. Hofmann Hofmann (1997) suggests that Ce/Pb values in the range 20–30 reflect melts of the primitive mantle, while samples with Ce/Pb <20 indicate contamination (Pb addition) from continental crust. (b) Ba/Th vs K/Nb variations of southeastern Ethiopia Plateau lavas. (c) Southeastern Ethiopian Plateau lavas have Ba/Nb vs La/Nb values within well-defined ranges of other suites from Ethiopia and northern Kenya. Sources for other data: HT1, HT2, LT (Pik *et al.*, 1998, 1999); Turkana (Furman *et al.*, 2004, 2006b); S. Ethiopia (George & Rogers, 2002); 23 Ma shield volcanoes (Choke and Guguftu, NW Ethiopia; Kieffer *et al.*, 2004); MER – Main Ethiopia Rift (Furman *et al.*, 2006a; Rooney *et al.*, 2007).

CaO/Al_2O_3 and Ni vs MgO; Figs 4 and 5). The data suggest that the Miocene Bale lavas originated and subsequently fractionated from two separate parental magmas and both parental magma compositions are distinct from those sampled at Chiro and the younger Bale lavas.

Crustal contamination

Lavas emplaced on the southeastern Ethiopian Plateau ascended through thick continental lithosphere (≥80 km, Dugda *et al.*, 2007), leading to potential interaction with the lithospheric mantle and continental crust. Samples with Ce/Pb <20 are candidates for crustal contributions, due to the high Pb content and low Ce/Pb ratio of continental crust. Seven out of nine Chiro lavas have Ce/Pb < 20 (Fig. 6); six of those samples form a subvertical trend towards decreasing Ce/Pb values, suggesting increasing Pb contributions. The lavas with Ce/Pb < 20 also show clear trends of increasing K/Nb and La/Nb without significant increases in Ba/Th or Ba/Nb, respectively. High K/Nb (>400) and La/Nb (>0·90) have been attributed to crustal contributions (Pik *et al.*, 1999; Kent *et al.*, 2002). In contrast, Chiro lava CH-02 has mantle-like Ce/Pb values due to lower Pb abundances. Chiro lava CH-09 has mantle Ce/Pb, La/Nb and K/Nb values, inconsistent with crustal contamination; CH-05, however, records borderline Ce/Pb (20·1) and K/Nb (404) suggesting crustal contributions may be possible. Isotopically, CH-09 has lower $^{87}Sr/^{86}Sr_i$ than CH-02 and CH-05, consistent with minimal crustal input (Fig. 8).

All Debre Sahil lavas have low Ce/Pb values (< 20) consistent with crustal contamination. Three samples consistently record high La/Nb, Ba/Nb, K/Nb and Ba/Th values (Fig. 6). However, samples DS-04 and DS-05 do not show extreme trace element ratios. They do, however, contain <1% alkali feldspar xenocrysts, indicating local interaction with crustal material. Isotopically, these two samples record ratios indicative of ancient incompatible element enrichment. However, their radiogenic Pb isotopic ratios do not trend towards crustal contributions (i.e. less radiogenic Pb), suggesting this signature may be caused by ancient mantle metasomatism (Fig. 8).

Of the Bale Mountains samples, some Pliocene Bale lavas have trace element signatures consistent with crustal contributions (Fig. 6). Specifically, BM-01 and BM-02 often plot with mildly contaminated Chiro samples and are geochemically distinct from other Pliocene samples. BM-07 and BM-23 record low Ce/Pb and elevated Ba/Th, K/Nb, Ba/Nb and La/Nb values, similar to contaminated Debre Sahil lavas. Because of their extreme trace element compositions and potential crustal contributions, none of these samples were analysed for their isotopic compositions.

Miocene Bale lavas have narrow ranges of Ba/Nb, La/Nb, K/Nb and Ba/Th values (Fig. 6), suggesting these trace element ratios are not helpful for evaluating

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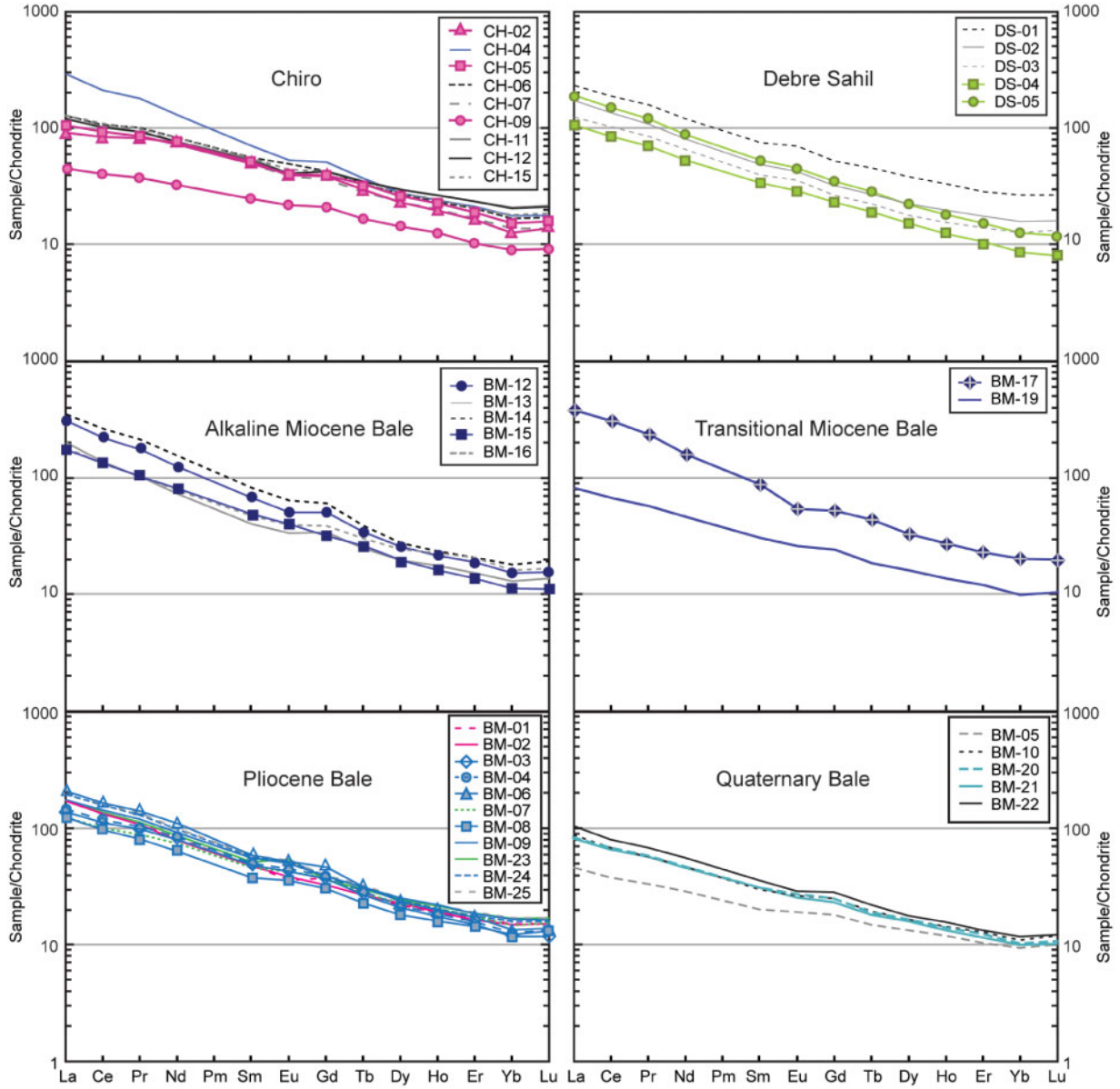


Fig. 7. Chondrite normalized REE variation diagrams. Chondrite values after Sun & McDonough (1989). In order to preserve the spacing associated with increasing atomic number and associated lanthanide contraction, Pm values were inferred but not measured.

Table 3: Helium analyses of southeastern Ethiopian Plateau lavas

Sample	wt analysed (mg)	3He/4He (R/R_A)	2σ	[He] ccSTP/g
CH-09	313·3	9·10	0·94	1·77E-9
BM-15	301·7	5·15	0·32	4·52E-9
BM-20	307·8	12·10	0·48	3·87E-9
BM-21	331·7	12·54	0·96	1·53E-9
DS-04	290·2	6·15	0·14	3·27E-9

crustal input in this suite. However, three alkaline samples (BM-13, BM-14 and BM-16) have Ce/Pb < 20, suggesting crustal input or metasomatic Pb addition. Consequently, these samples were not analysed isotopically. Basal alkaline Miocene Bale lava (BM-12) contains alkali feldspar xenocrysts, indicating local interaction with crustal material. BM-12 is intermediate isotopically (Sr–Nd–Pb–Hf–Os) between the other alkaline lava (BM-15) and the two transitional lavas. This may indicate crustal contamination or a metasomatic contribution between that of the other lavas.

Osmium isotopic ratios can be extremely sensitive to small amounts of radiogenic crustal input ($^{187}Os/^{188}Os$ ~1; Esser & Turekian, 1993), especially when magmatic Os concentrations are low. Samples BM-12 and DS-04 have radiogenic $^{187}Os/^{188}Os_i$ ratios (0·1623 and 0·1757, respectively) compared to primitive mantle (0·1296; Meisel *et al.*, 2001), possibly due to crustal interaction. Sample BM-12 has a low Os content (5 ppt); the higher Os content of DS-04 (195 ppt) can be attributed to olivine accumulation. Other southeastern Ethiopian

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Table 4: Southeastern Ethiopian Plateau Sr–Nd–Pb–Hf isotopic data

Sample	$^{87}Sr/^{86}Sr$	2SE	$^{143}Nd/^{144}Nd$	2SE	ε Nd	$^{176}Hf/^{177}Hf$	2SE	ε Hf	$^{206}Pb/^{204}Pb$	2SE	$^{207}Pb/^{204}Pb$	2SE	$^{208}Pb/^{204}Pb$	2SE
CH-02	0·704221	4	0·512887	5	4·85	0·283082	4	10·98	18·8962	7	15·5755	7	38·5697	17
CH-05	0·704354	7	0·512846	5	4·07	0·283037	4	9·35	18·5056	8	15·5729	6	38·9114	16
CH-05 (2)			0·512844	6	4·02									
CH-09	0·703938	1	0·512881	5	4·73	0·283054	5	9·99	18·5052	7	15·5542	6	38·4875	17
BM-03	0·703827	13	0·512753	5	2·24	0·282955	4	6·46	18·5195	6	15·5848	6	38·5503	17
BM-04	0·703781	13	0·512753	5	2·25	0·282962	9	6·71	18·5438	4	15·5888	4	38·5872	12
BM-06	0·703726	11	0·512777	4	2·70	0·282964	7	6·81	18·8621	9	15·5966	9	38·8207	26
BM-08	0·703897	2	0·512742	6	2·02	0·282937	5	5·83	18·4840	4	15·5858	4	38·5432	9
BM-09	0·703796	4	0·512749	6	2·17	0·282970	4	7·01	18·7450	6	15·5913	5	38·7683	20
BM-09 (2)			0·512761	7	2·41									
BM-12	0·703342	19	0·512744	5	2·06	0·282871	4	3·49	19·7867	7	15·6514	6	39·5958	17
BM-15	0·703642	15	0·512708	4	1·37	0·282849	11	2·72	20·0742	5	15·7125	5	39·9335	11
BM-17	0·703410	1	0·512804	4	3·23	0·282943	5	6·06	19·2701	6	15·6191	5	39·2646	15
BM-19	0·703532	5	0·512819	5	3·52	0·282965	6	6·83	19·1600	9	15·6093	6	39·2003	17
BM-20	0·703493	11	0·512851	5	4·15	0·282987	5	7·59	18·8876	7	15·5830	6	38·9243	17
BM-21	0·703425	4	0·512851	4	4·16	0·282990	5	7·72	18·8436	11	15·5775	14	38·9029	23
BM-24	0·703798	7	0·512758	5	2·33	0·282981	5	7·40	18·7898	7	15·5908	6	38·7560	16
DS-04	0·704091	14	0·512662	4	0·47	0·282816	4	1·55	19·2736	7	15·6057	5	39·4672	16
DS-05	0·703915	14	0·512764	4	2·46	0·282888	4	4·12	19·7307	6	15·6599	6	39·8261	17

Measured values of Sr, Nd, Pb and Hf isotopes. The reported uncertainties are 2 standard errors of the mean (2SE) for the in-run precision and thus provide a measure of sample reproducibility. The NIST SRM 987 standard averaged $^{87}Sr/^{86}Sr = 0{\cdot}710233 \pm 0{\cdot}000007$ 2SE, n = 4, AMES Nd standard averaged $^{143}Nd/^{144}Nd = 0{\cdot}512132 \pm 0{\cdot}000002$ 2SE, n = 22, JMC 475 Hf standard averaged $0{\cdot}282185 \pm 0{\cdot}000020$ 2SE, n = 22 and NIST NBS SRM 981 averaged $^{206}Pb/^{204}Pb = 16{\cdot}9430 \pm 0{\cdot}0010$, $^{207}Pb/^{204}Pb = 15{\cdot}5003 \pm 0.0004$, $^{208}Pb/^{204}Pb = 36{\cdot}7270 \pm 0{\cdot}0004$ 2SE, n = 15.

Plateau lavas also have low Os concentrations and radiogenic $^{187}Os/^{188}Os_i$. Quaternary Bale and MER lavas show increasing $^{187}Os/^{188}Os_i$ with decreasing Os concentration, which is often a hallmark of crustal contamination. However, MgO-rich and Os-rich (56–1454 ppt) magmas at Toro Ankole ($^{187}Os/^{188}Os = 0{\cdot}1475$–$0{\cdot}4974$; Rosenthal *et al.*, 2009) and Turkana ($^{187}Os/^{188}Os_i = 0{\cdot}1450$–$0{\cdot}1480$; Nelson *et al.* 2012) demonstrate that there is a significant regional variability in mantle Os isotopic signatures. The negative correlation between $^{187}Os/^{188}Os_i$ and K/Nb in Miocene Bale lavas suggests the increasing radiogenic signature is not due to crustal contributions (Fig. 10). The lack of correlation between Ce/Pb and $^{87}Sr/^{86}Sr$ also argues against crustal contamination. Instead, positive $^{187}Os/^{188}Os_i$ – Ba/Rb and $^{187}Os/^{188}Os_i$ – $^{206}Pb/^{204}Pb_i$ suggest ancient metasomatism is responsible for the radiogenic $^{187}Os/^{188}Os_i$ ratios. DS-4 contains xenocrysts and has elevated $^{187}Os/^{188}Os_i$ and higher K/Nb, indicating continental crust likely contributed to the parental magma of this sample. The correlation of radiogenic Os isotopes with high K/Nb in CH-02 confirms crustal contributions. Therefore, crustal contributions do influence some of the Os isotope systematics, but cannot account for all of the radiogenic values, especially in the Bale Mountains lavas.

Depths of partial melting and source mineralogy

A major goal of this study is to understand the evolving thermal and mineralogical structure of the lithosphere during the transition from initial mantle plume volcanism to the eruption of MORB-like lavas that characterize the rifting environment. This requires quantification of the depth of melt generation, plus the mineralogic and isotopic makeup of the source regions beneath the southeastern Ethiopian Plateau during the period of post-flood basalt volcanism. The following discussion considers trace element covariations as proxies for lithospheric thickness because they carry information about how potential source mineral assemblages varied with depth, both spatially and temporally. Specifically, REE ratios can be used as a proxy to estimate the depth of melting with respect to the garnet–spinel transition zone, because garnet strongly partitions HREE.

The melting depth and thus the inferred thickness of the lithospheric lid appears to have varied systematically across the region. Among the Oligocene–Miocene basalts, the depth of melting increased from north to south across the plateau. Chiro lavas have Tb/Yb_N and Dy/Yb_N values (1·9–2·3 and 1·4–1·8, respectively) that suggest melting occurred near the spinel–garnet transition, extending into the garnet field (Fig. 11; Wang *et al.*, 2002). Coeval Debre Sahil (DS-04 and DS-05) and alkaline Miocene Bale lavas have overlapping ratios of Tb/Yb_N (2·3–2·4), Dy/Yb_N (1·5–1·7) and La/Yb_N (15·3–21·8), suggesting they originated from similar depths of melting. Transitional Miocene Bale lavas (BM-17 and BM-19; La/Yb_N 10·3–13·7, Tb/Yb_N 2·5–2·9, Dy/Yb_N 1·9–2·2; Fig. 11), record the highest ratios and, by extension, the greatest depth of melting. In contrast, Pliocene and Quaternary Bale lavas have lower Tb/Yb_N (1·8–2·3) and Dy/Yb_N (1·4–1·7) values than the Miocene Bale lavas and suggest melting at shallower depths near the garnet–spinel transition zone.

MREE/HREE ratios can be influenced by amphibole and pyroxene in hydrated or veined peridotite,

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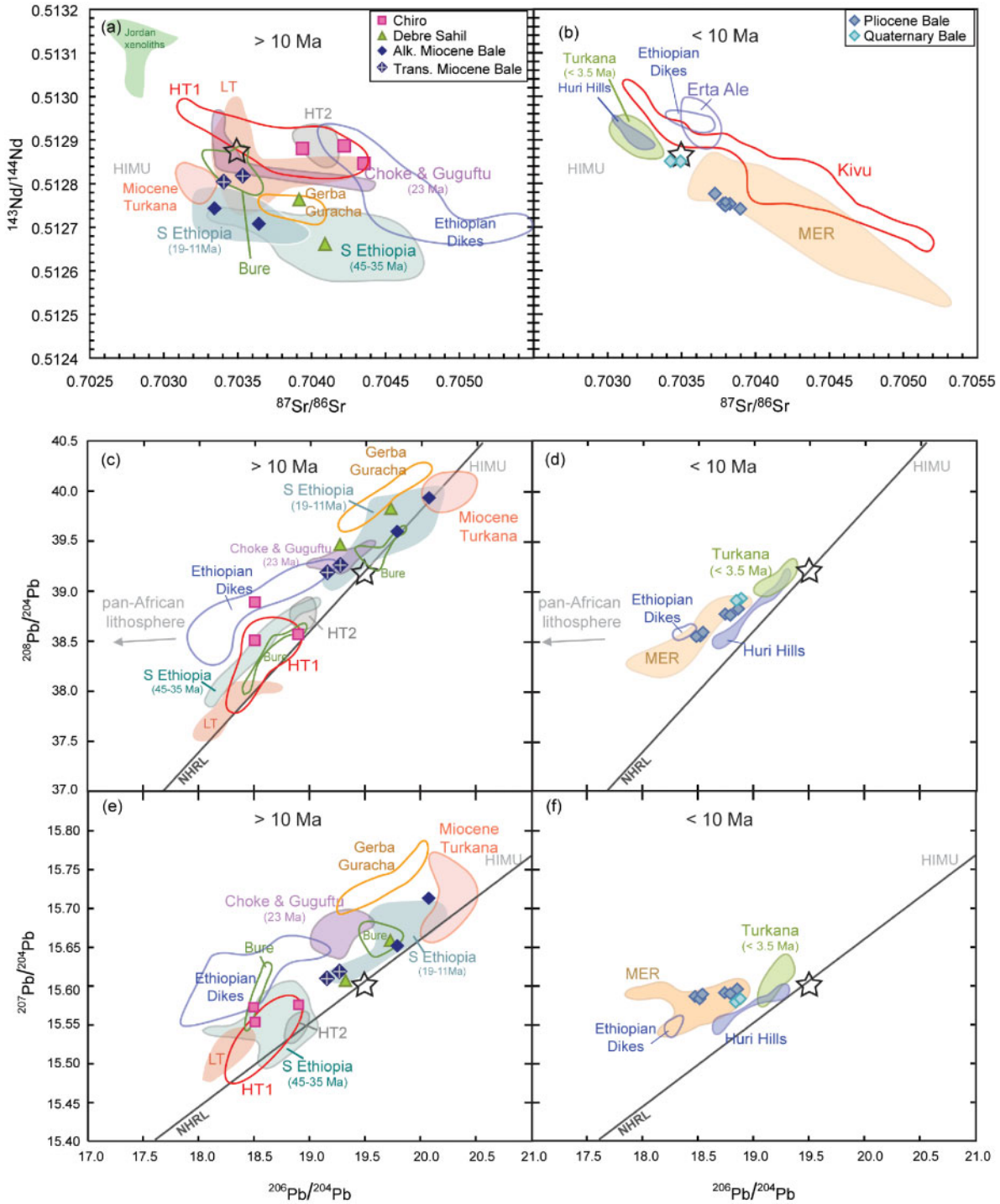


Fig. 8. Initial Sr–Nd–Pb isotopic data for southeastern Ethiopian Plateau lavas, presented in groups based on lava age (>10 Ma and < 10 Ma). Data for other Afro-Arabian volcanic localities include: the Afar plume (star; defined by Rooney *et al.*, 2012); Jordan xenoliths (Shaw *et al.*, 2007); HT1-HT2-LT basalts (Pik *et al.*, 1999; Beccaluva *et al.*, 2009); Turkana (Furman *et al.*, 2004, 2006b); Huri Hills (Class *et al.*, 1994); Choke and Guguftu volcanoes (Kieffer *et al.*, 2004); southern Ethiopia (George *et al.*, 1998, George & Rogers, 2002); Bure (Meshesha & Shinjo, 2007); Erta Ale (Barrat *et al.*, 1998); Main Ethiopian Rift (MER; Hart *et al.*, 1989; Furman *et al.*, 2006a; Rooney *et al.*, 2012); Ethiopian Rift margin dikes (Rooney *et al.*, 2013); Gerba Guracha (Rooney *et al.*, 2014). Northern Hemisphere Reference Line (NHRL Hart, 1984).

respectively. These minerals preferentially sequester the MREE and, therefore, can exert control over Tb and Dy. Partial melting of amphibole-bearing peridotite, for example, can generate a MREE concavity in chondrite-normalized REE plots. The parameter Dy/Dy* enables quantification of chondrite-normalized REE plots (Davidson *et al.*, 2013). Low degrees of partial melting in the presence of garnet will not significantly increase

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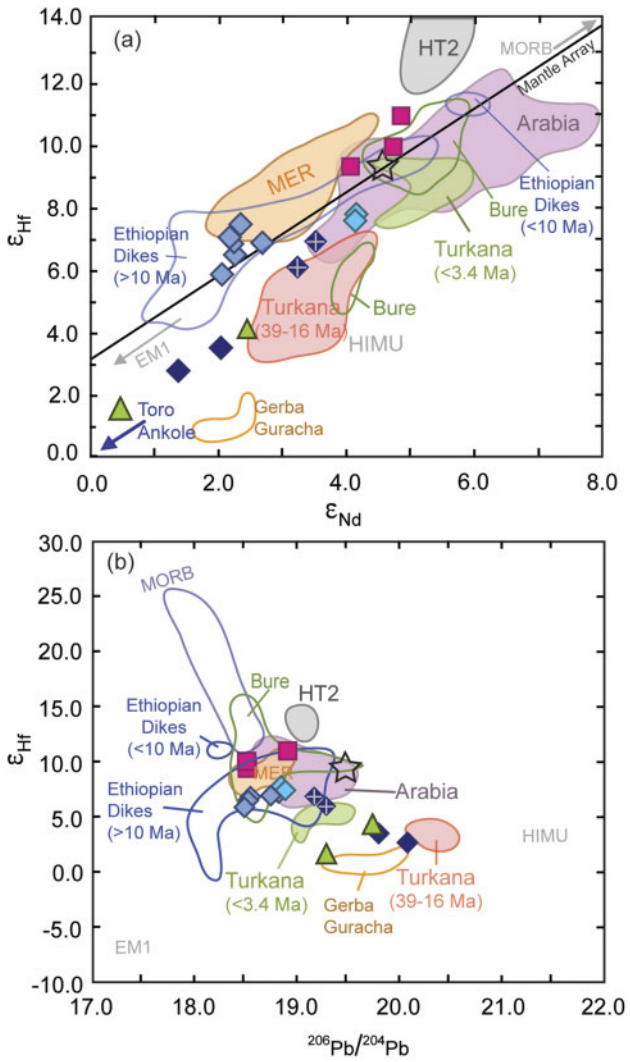


Fig. 9. A summary of initial Hf and Nd isotopic data available for East African Rift lavas, including southeastern Ethiopian Plateau lavas. (a) ε_{Hf} vs ε_{Nd} data and (b) ε_{Hf} vs $^{206}Pb/^{204}Pb$ data. Fields for other volcanic localities include the Afar plume (star; defined by Rooney *et al.* 2012), Toro Ankole (Rosenthal *et al.*, 2009), Turkana (Locke, 2008; Locke *et al.*, 2008), Bure (Meshesha & Shinjo, 2007; 2010), Saudi Arabia (Bertrand *et al.*, 2003), Ethiopian Rift margin dikes (Rooney *et al.*, 2013), and Gerba Guracha (Rooney *et al.*, 2014). The MORB field is from Salters & White (1998).

Dy/Dy* (i.e. it will be ~1·0), but it will generate elevated Dy/Yb ratios (Fig. 11). Residual amphibole can sequester MREE, thereby leading to lower Dy/Dy* and Dy/Yb values. The presence of clinopyroxene does not influence Dy/Yb values significantly, but can lead to lower Dy/Dy*. Assimilation of middle and upper continental crust, which is dominantly made during subduction processes that often involve the crystallization and fractionation of amphibole, can likewise lead to decreased Dy/Dy* and Dy/Yb values.

Consistent with the previous observation, transitional Miocene Bale lavas and three Chiro lavas (CH-02, CH-05 and CH-09) have high Dy/Dy*–Dy/Yb values indicative of residual garnet in the mantle source (Fig. 11). Despite elevated Dy/Yb and Tb/Yb_N values, alkaline Miocene Bale lavas and two Debre Sahil lavas (DS-04 and DS-05) record low Dy/Dy*, suggesting the presence of residual clinopyroxene and the absence of garnet in the source. Crustally contaminated Chiro and Debre Sahil lavas also show trends to lower Dy/Dy* and Dy/Yb, consistent with involvement of middle and upper continental crust. Notably, Pliocene and Quaternary Bale lavas do not show evidence for significant crustal contamination and, therefore, their trend towards low Dy/Dy* and Dy/Yb values suggests residual amphibole is present in the source, potentially at shallower lithospheric mantle depths. These geochemical indicators appear to require significant temporal variation in the mineralogy and depth of melting, potentially reflecting local thinning of the lithosphere over time.

Mineralogical variability in the lithosphere is significantly influenced by its metasomatic history (e.g. carbonatite, hydrous, low-volume silicate melt) and can be identified through the bulk geochemistry of mafic lavas (e.g. Baker *et al.*, 1997; George & Rogers, 2002; Beccaluva *et al.*, 2009; Rosenthal *et al.*, 2009; Rooney *et al.*, 2014). Direct evidence for regional lithospheric metasomatism comes from xenoliths sampled in widely separated portions of the western Ethiopian Plateau, the Arabian Peninsula, western Uganda, northern Tanzania and the Ethiopia/Kenya border region (e.g. Davies & Lloyd, 1986; Dawson & Smith, 1988; Henjes-Kunst & Altherr, 1992; Rudnick *et al.*, 1993; Blusztajn *et al.*, 1995; Paslick *et al.*, 1995; Baker *et al.*, 1998; Chesley *et al.*, 1999; Burton *et al.*, 2000; Dawson, 2002; Shaw *et al.*, 2007; Rooney *et al.*, 2017a; Trestrail *et al.*, 2017). In particular, minor amphibole in spinel lherzolite and harzburgite xenoliths in Ethiopia and Yemen (Baker *et al.*, 1998; Conticelli *et al.*, 1999; Ferrando *et al.*, 2008; Beccaluva *et al.*, 2011) demonstrate that the local upper lithospheric mantle contains sufficient water to undergo partial melting at temperatures lower than those required for anhydrous lithosphere. Pyroxenite xenoliths sampled on the western Ethiopian Plateau attest to the presence of a pyroxenite underplate in some regions of Ethiopia (Rooney *et al.*, 2017a).

Differences in the mineralogy of the mantle and lithospheric sources contributing to southeastern Ethiopian Plateau magmatism are recorded in the trace element abundance patterns of the lavas. Here, trace element data are compared qualitatively to numerical models of melts derived from variably metasomatized source materials (Pilet *et al.*, 2011). Chiro lavas (excluding the 28 Ma basal lava CH-02) have broadly smooth primitive mantle-normalized trace element patterns consistent with their derivation by <1% melting of an anhydrous source such as peridotite or anhydrous veins of clinopyroxene in the mantle (Pilet *et al.*, 2011; Fig. 12). Chiro lavas have uniform 10 000*Zn/Fe values (~12; Fig. 13), further suggesting melting of an anhydrous peridotite source at depths close to the garnet–spinel transition zone. CH-09 has accumulated olivine, but olivine fractionation does not influence the 10 000*Zn/Fe ratios

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Table 5: Re–Os isotopes and abundances for selected lavas from the Southeastern Ethiopian Plateau

Sample	MgO (wt %)	$^{187}Os/^{188}Os_m$	2SE	$^{187}Os/^{188}Os_i$	$^{187}Re/^{188}Os_m$	Os (ppt)	Re (ppt)
Southeastern Ethiopian Plateau (this study)							
CH-09	10·38	0·1898	0·00024	0·1813	33·03	51	349
BM-12	6·14	0·5497	0·00181	0·1625	893·9	5	930
BM-15	9·32	0·1883	0·00035	0·1754	31·03	31	199
BM-17	9·88	0·1651	0·00121	0·1484	40·05	18	151
BM-20	9·91	0·2268	0·00061	0·2266	17·79	12	44
DS-04	14·33	0·1785	0·00008	0·1757	6·97	195	281
Quaternary Main Ethiopian Rift (Furman et al., 2006b; Rooney et al., 2007)							
N-15	*8·76*	0·1797	0·00127	0·1787	61·68	7	93
N-18	*7·48*	0·1876	0·00090	0·1804	432·99	3	268
N-21	*10·53*	0·1515	0·00029	0·1505	62·27	25	314
N-22	*8·09*	0·2293	0·00030	0·2286	40·90	11	94
Pliocene Turkana (Furman et al., 2004)							
KCH-18	*8·64*	0·1761	0·00037	0·1726	142·42	10	294
SI-24	*11·57*	0·2360	0·00009	0·2356	14·83	134	406
CI-10A	*5·64*	0·1595	0·00018	0·1576	78·11	20	330

For comparison, we also analysed select Quaternary lavas from the Main Ethiopian Rift (Furman *et al.*, 2006a; Rooney *et al.*, 2007) and Pliocene lavas from Turkana (Furman *et al.*, 2004); the MgO values in italics have been reported previously in the cited works. Uncertainties on the data are expressed as 2SE (standard error of the mean).

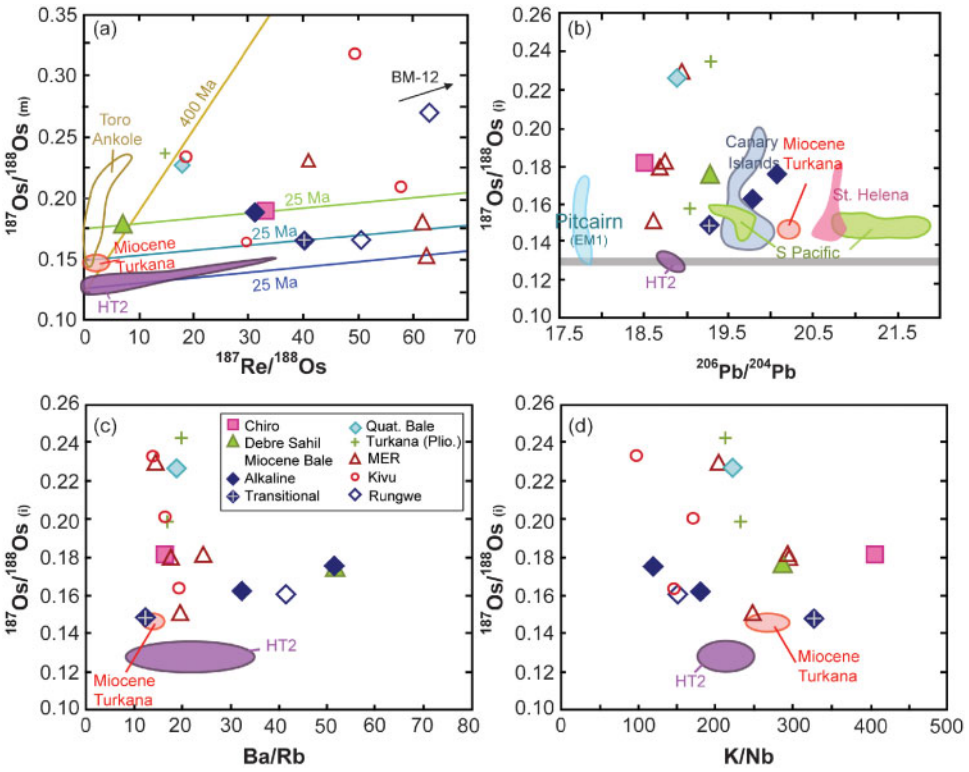


Fig. 10. (a) Measured $^{187}Os/^{188}Os$ vs $^{187}Re/^{188}Os$ data for selected southeastern Ethiopian Plateau lavas. The age reference lines were calculated using the ages of the lavas and the HT2 and HIMU Turkana end-members reported for the EARS. Sources of data for other EARS lavas: HT2 flood basalts (Rogers *et al.*, 2010; Nelson *et al.*, 2012); Toro Ankole (Rosenthal *et al.*, 2009); Miocene Turkana, Kivu and Rungwe (Nelson *et al.*, 2012). (b) Initial $^{187}Os/^{188}Os$ vs $^{206}Pb/^{204}Pb$ systematics. The $^{187}Os/^{188}Os$ composition of primitive mantle is shown as a grey bar (Meisel *et al.*, 2001). Additional sources of data: St. Helena (Reisberg *et al.*, 1993); Pitcairn (Eisele *et al.*, 2002); southern Pacific (Hauri & Hart, 1993); Canary Islands (Marcantonio *et al.*, 1995). (c) $^{187}Os/^{188}Os_i$ vs Ba/Rb and (d) $^{187}Os/^{188}Os_i$ vs K/Nb ratios for southeastern Ethiopian Plateau lavas.

(Le Roux *et al.*, 2010) and, therefore, is considered a peridotitic melt. The trace element data suggest Chiro lavas were derived from a dominantly anhydrous, garnet-bearing peridotite, potentially containing metasomatic pyroxenite veins (Le Roux *et al.*, 2010, Ducea *et al.*, 2013).

In contrast, the primitive mantle-normalized trace element abundance patterns of the basal Chiro lava (CH-02) plus the Debre Sahil lavas and alkaline Miocene Bale basalts suggest melting of a mixed cumulate assemblage containing significant amphibole +

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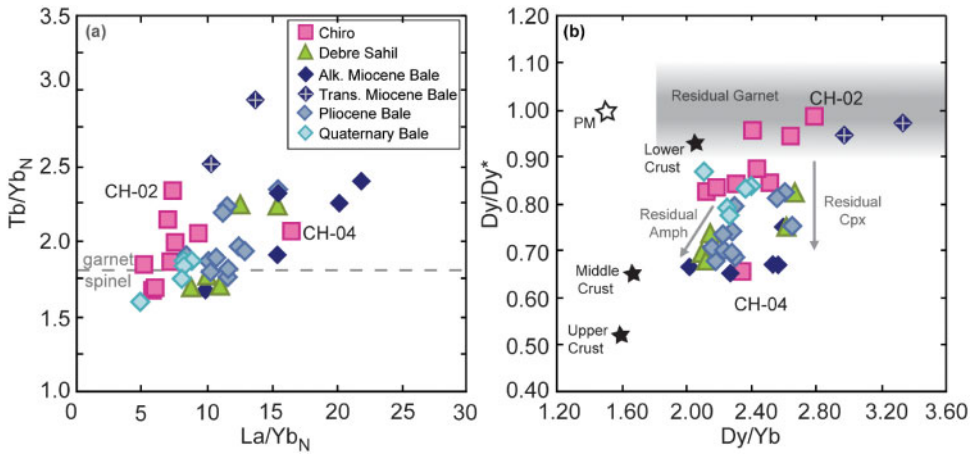


Fig. 11. (a) Tb/Yb$_N$ vs La/Yb$_N$ values ('N' indicates chondrite normalized values). Tb/Yb$_N$ >1·8 suggests melting in the presence of garnet or abundant pyroxene (Wang *et al.*, 2002). CH-04 is a more evolved sample (MgO = 1·76 wt %) and was not used to evaluate depth of melting. (b) Dy/Dy* vs Dy/Yb values to evaluate crustal input and the presence of residual minerals in the mantle. Davidson *et al.* (2013) summarized endmember compositions and provided the basis for Dy/Dy* calculations.

clinopyroxene (up to 50%) ± minor orthopyroxene, garnet and phlogopite ('hydrous cumulate assemblage' of Pilet *et al.*, 2011; Fig. 12). Alkaline Miocene Bale lava BM-15 has 10 000*Zn/Fe ~25 and Dy/Dy* = 0·75, suggesting abundant pyroxenite in its mantle source. In contrast, Debre Sahil basalts and the remaining alkaline Miocene Bale lavas straddle the pyroxenite–peridotite divide (10 000*Zn/Fe ~12) but lack evidence for garnet in the source (Fig. 13). This suggests these lavas were formed at shallower levels and/or preferentially melted garnet-free metasomatic veins.

Transitional Miocene Bale lavas also have high Zn/Fe ratios, but the Dy/Dy* values indicate residual garnet in the source (Fig. 13). Like clinopyroxene, garnet can fractionate Zn from Fe, generating high Zn/Fe ratios and precluding lithology identification (i.e. garnet peridotite versus garnet pyroxenite) using these indicators. Transitional Miocene Bale lavas lack the pronounced Ba and Nb enrichments (relative to U-Th depletions) of lavas derived from an amphibole-bearing source and, instead, likely originated from a dominantly anhydrous peridotite mantle source that may have contained veins of low-volume silicate melt (Fig. 12). The greater depth of melting recorded by the REE ratios also suggests these lavas originated at pressures greater than the amphibole stability field.

Pliocene and Quaternary Bale lavas have trace element patterns similar to those of amphibole within veins of metasomatized continental lithospheric mantle (Fig. 12; *cf* Pilet *et al.*, 2011); Zn/Fe values for these lavas suggest mostly peridotitic contributions (Fig. 13) and their REE abundances indicate melting occurred near the garnet–spinel transition zone or in the spinel stability field (Fig. 7). Subtle differences in trace element patterns (in particular, minor Ti depletion; Fig. 12) suggest a role for residual Ti oxides in both the young Bale lavas and the basal Chiro lava (CH-02). Pliocene Bale lavas show more pronounced trace element variations than the Quaternary Bale lavas and, in particular, have higher La/Nb, Ba/Nb and Rb/Sr values; we interpret this sequence as recording Pliocene low-degree melting of an amphibole-bearing peridotite source, followed by Quaternary higher-degree melting of the now residual source.

Isotopic signatures of source domains for Ethiopian Plateau volcanism

Metasomatic signatures in xenoliths and lavas

The Sr–Nd–Pb–Hf–He–Os isotope compositions of the southeastern Ethiopian Plateau volcanic suites cannot be explained solely by melting of Afar plume material (Rooney *et al.*, 2012). Instead, the style and timing of metasomatism in the region varied both spatially and temporally, making each volcanic suite unique. Chiro lavas are similar to HT1 flood basalts, both compositionally (major and trace elements) and isotopically, consistent with derivation by melting of lithospheric mantle with contributions from the Afar plume (Pik *et al.*, 1999; Furman *et al.*, 2016). All other Oligocene–Miocene lavas reported here have isotopic analogues in the flood and plateau basalts from central and southern Ethiopia (LT flood basalts, southern Ethiopia, northern Kenya; Figs 8 and 9). Those lavas also incorporated melts of metasomatized lithosphere or continental crust (e.g. Stewart & Rogers, 1996; Pik *et al.*, 1999; Rooney *et al.*, 2014, 2017b). Pliocene Bale basalts have isotopic characteristics that resemble modern lavas from the Main Ethiopian Rift (Rooney *et al.*, 2012), while the younger Quaternary Bale suite has Sr–Nd–He (but not Pb–Hf) isotopic compositions that are similar to the modern Afar plume. These isotopic features are consistent with the trace element evidence for the involvement of variably metasomatized lithosphere.

Three styles of mantle metasomatism have been inferred from the mineralogy and geochemistry of lavas and mantle xenoliths across Afro-Arabia: (1) volatile-rich, subduction-related metasomatism from the

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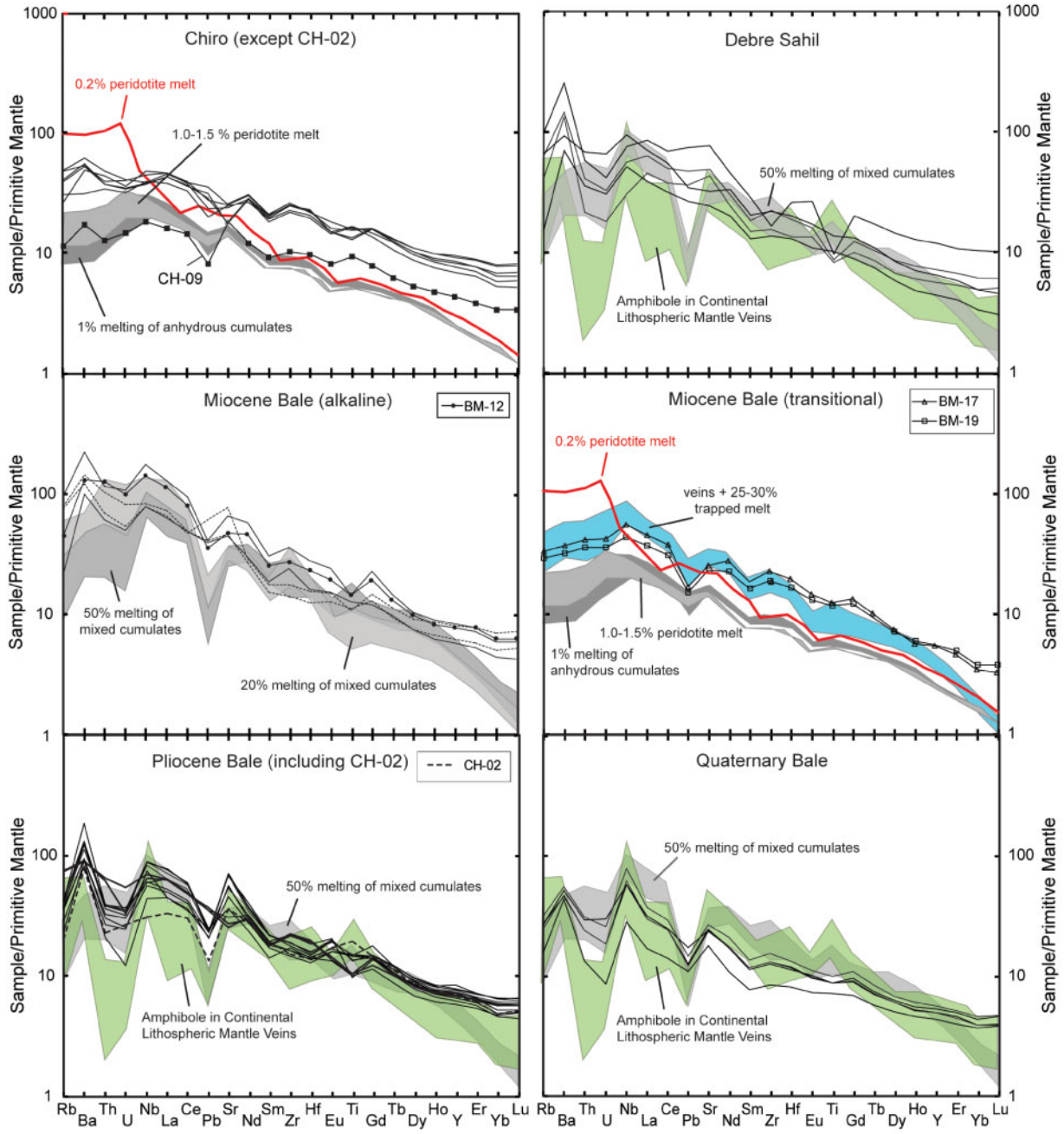


Fig. 12. Primitive mantle normalized incompatible trace element patterns for each suite. Sample CH-02 (28 Ma) is plotted with Pliocene Bale samples to highlight their similar trace element patterns. Mass balance trace element models from Pilet *et al.* (2011) were used to determine the trace element budgets of melt and subsequent differentiated lavas produced from hydrous or anhydrous mantle lithologies. Fields for best-fit models are included for comparison with samples in this study. 'Mixed cumulates' is the equivalent of 'hydrous cumulates' in Pilet *et al.* (2011) as the assemblage contains both hydrous and anhydrous minerals (amphibole + up to 50% clinopyroxene ± minor orthopyroxene, garnet and phlogopite). Primitive mantle values are from McDonough & Sun (1995).

Pan-African orogeny (500–900 Ma) or an earlier subduction event (e.g. Stein *et al.*, 1997; Frezzotti *et al.*, 2010; Rooney *et al.*, 2014); (2) carbonatite metasomatism associated with either ancient subduction or more recent plume activity (e.g. Rudnick *et al.*, 1993; Rosenthal *et al.*, 2009; Trestrail *et al.*, 2017); and (3) alkaline silicic metasomatism associated broadly with the Afar plume (e.g. Dawson & Smith, 1988; Rogers *et al.*, 1992; Bedini *et al.*, 1997; Stein *et al.*, 1997; Baker *et al.*, 1998; Chesley *et al.*, 1999; Spath *et al.*, 2001; Kaeser *et al.*, 2006; Beccaluva *et al.*, 2009; Sgualdo *et al.*, 2015).

The xenolith population sampling the Ethiopian lithospheric mantle is dominated by spinel peridotites, with traces of metasomatic amphibole (e.g. Conticelli *et al.*, 1999; Ferrando *et al.*, 2008; Frezzotti *et al.*, 2010; Alemayehu *et al.*, 2017; Trestrail *et al.*, 2017). These

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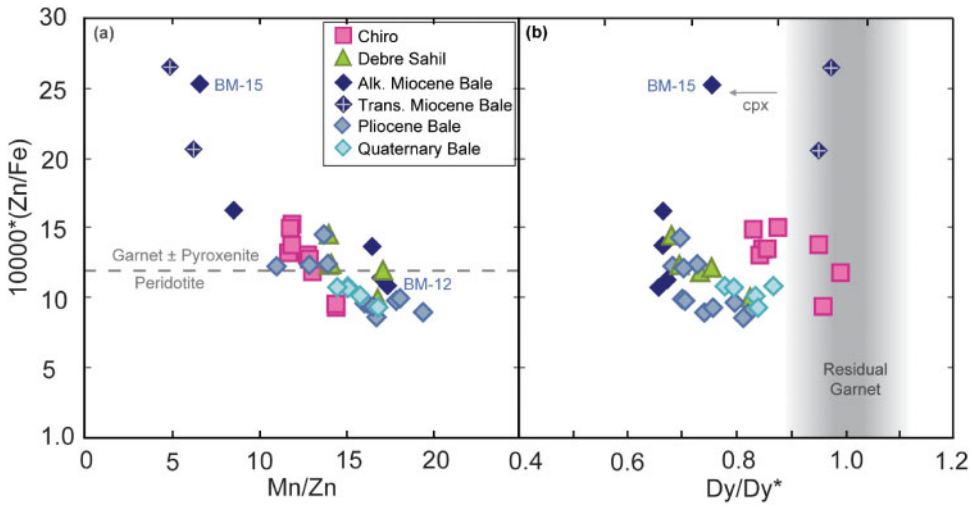


Fig. 13. (a) The partitioning behavior of Zn and Fe in peridotite-derived melts results in 10 000*Zn/Fe <12, whereas melts derived from a garnet-bearing or pyroxenite source will have ratios >12. (b) 10 000*Zn/Fe vs Dy/Dy* values for southeastern Ethiopian Plateau lavas. Field for residual, garnet-bearing sources is from Davidson *et al.* (2013).

xenoliths have unradiogenic $^{87}Sr/^{86}Sr$, radiogenic $^{143}Nd/^{144}Nd$ and $^{176}Hf/^{177}Hf$ and variable $^{206}Pb/^{204}Pb$ (Blusztajn *et al.*, 1995; Shaw *et al.*, 2007; Beccaluva *et al.*, 2011; Nelson *et al.*, 2011), whereas the crust generally has high $^{87}Sr/^{86}Sr$ and low $^{143}Nd/^{144}Nd$ and $^{206}Pb/^{204}Pb$ (Baker *et al.*, 1996b). Metasomatized 'Pan-African lithosphere' has sometimes been invoked to explain lava isotopic signatures that are quite unlike those of the mantle xenoliths, e.g. unradiogenic $^{143}Nd/^{144}Nd$, $^{176}Hf/^{177}Hf$, $^{206}Pb/^{204}Pb$ and radiogenic $^{87}Sr/^{86}Sr$. These characteristics cannot be readily explained by crustal contamination and are instead attributed to interaction with a metasomatized mantle source (e.g. Schilling *et al.*, 1992; Endress *et al.*, 2011; Rooney *et al.*, 2012, 2014). These disparate isotopic signatures may indicate that the upper levels of the lithospheric mantle are isotopically distinct from the lower levels of the metasomatized 'Pan-African lithosphere'. For example, pyroxenite xenoliths from the western Ethiopian Plateau are thought to be cumulates that experienced significant interaction with melts formed in association with the 30 Ma flood basalts (Rooney *et al.*, 2017a). These observations suggest that isotopic variations in southeastern Ethiopian Plateau lavas did not originate strictly from dry and depleted shallow lithospheric mantle, but may instead be derived from additional mantle components that have not yet been observed in the sampled xenoliths.

HIMU isotopic signatures

The HIMU-trending Pb isotopic signatures observed in Miocene Debre Sahil and alkaline Bale lavas are found in Oligocene–Miocene volcanic suites across Afro-Arabia and are associated consistently with evidence of contributions from metasomatized mantle material. In Ethiopia and northern Kenya, this signature is restricted primarily to Oligocene and early Miocene lavas. The earliest occurrence is in the 35–39 Ma Mt. Porr and Kajong lavas of Turkana, northern Kenya (Furman *et al.*, 2006b), while the next is in 28 Ma Bure lavas (Meshesha & Shinjo, 2007) and 24–25 Ma lavas in Gerba Guracha in northwest Ethiopia (Rooney *et al.*, 2014, 2017b); the youngest occurrence is at Getra Kele, S Ethiopia (12 Ma; George *et al.*, 1998, George & Rogers, 2002). In addition, primitive mafic lavas that record a range of HIMU contributions are found sporadically along the EARS (this study; Class *et al.*, 1994; George & Rogers, 2002; Kieffer *et al.*, 2004; Meshesha & Shinjo, 2007).

There are two geologically reasonable models that can account for the widespread appearance of HIMU-trending mafic lavas in the EARS from the Oligocene to the mid-Miocene. First, this characteristic signature could be a broad compositional feature located within the lower lithospheric mantle that underlies much of the EARS. Numerous studies have argued that local to regional metasomatic activity created chemically heterogeneous domains in the continental lithosphere (e.g. Spath *et al.*, 2001; Bertrand *et al.*, 2003; Shaw *et al.*, 2003; Locke *et al.*, 2008; Tesfaye *et al.*, 2008; Beccaluva *et al.*, 2009; Rooney *et al.*, 2014). Melting of this lithospheric layer produced mafic melts with HIMU-like isotopic values for a restricted period of time. Second, the HIMU component could reside in sublithospheric packages of variably metasomatized recycled oceanic lithosphere entrained within the broad, upwelling South African Superplume and associated plume-derived fluids (e.g. Bell & Simonetti, 1996; Chesley *et al.*, 1999; Furman & Graham, 1999; Furman *et al.*, 2006b; Geldmacher *et al.*, 2011; Meshesha & Shinjo, 2007, 2008; Aulbach *et al.*, 2008; Aulbach & Rudnick, 2009; Nelson *et al.*, 2012; Natali *et al.*, 2016). Near-surface mantle upwelling extending beneath Afar, Kenya and Tanzania reflects, directly or indirectly, the thermal impact of the superplume (Ni *et al.*, 2002; Simmons *et al.*, 2007; Hansen *et al.*, 2012; Bagley & Nyblade, 2013) and, consequently, the geochemical variations observed

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across the EARS may reflect heterogeneities fundamental to this large, deep-seated thermochemical feature (Furman *et al.*, 2006b; Nelson *et al.*, 2012). Both models are evaluated with respect to southeastern Ethiopian Plateau lavas in light of the available geochemical and geophysical data from the EARS.

Model 1: Lithospheric controls on Ethiopian Plateau volcanism

There is clear geochemical evidence that the lithospheric mantle contributes to southeastern Ethiopian Plateau volcanism. The strong HIMU signature in alkaline Miocene Bale and Debre Sahil lavas correlates with trace element evidence for metasomatic contributions (Figs 6 and 10) and provides compelling evidence that the HIMU mantle component is located within the lithosphere beneath Ethiopia. Trace element abundances in these lavas (Fig. 12; Pilet *et al.*, 2011) require a hydrated source (i.e. amphibole veins ± minor phlogopite) that likely contains secondary clinopyroxene and melt trapped in the surrounding peridotite. Trace element evidence indicates the majority of HIMU-like mafic lavas have negative primitive-mantle normalized K, U and Th anomalies that suggest melting of an amphibole-bearing metasome (e.g. Furman *et al.*, 2006b; Rooney *et al.*, 2014, 2017b). In this study, samples with the highest $^{206}Pb/^{204}Pb$ ratios also show the largest negative primitive mantle-normalized K anomalies; transitional Miocene Bale lavas lack these geochemical and isotopic correlations and must be derived from a mineralogically distinct mantle source. Uncontaminated Pliocene Bale lavas show small K depletions, whereas Quaternary Bale lavas, which are generated by a higher degree of melting, lack K anomalies.

Combined trace element and isotopic variations enable us to identify at least two episodes of metasomatic activity beneath the southeastern Ethiopian Plateau: an older metasomatic event affecting the lithosphere beneath Debre Sahil and Bale (likely associated with the Pan-African orogeny) was followed by a much younger and shallower metasomatic event in Bale and, potentially, to the north in Chiro. The timing of the latter event is constrained only to precede eruption of the ~18·5 Ma Chiro basalts and thus may be associated with the Afar plume.

The thickness of the lithosphere and the inferred depth of melting beneath the EARS are both highly variable through space and time, making it difficult to envision a continuous lithospheric layer that contributes to volcanism both on the thick Ethiopian Plateau and in the highly-extended Turkana depression (Keller *et al.*, 1994; Dugda *et al.*, 2007). The HIMU-trending signature may record a process tied directly to ancient (>500 Ma) metasomatism of continental or oceanic lithosphere (e.g. Ballentine *et al.*, 1997; Stracke *et al.*, 2003). Hydrous metasomes will evolve via radiogenic ingrowth to manifest a HIMU-trending signature within ~500 Myr of the metasomatic event; the intensity of the HIMU signature is dependent on the degree of metasomatism and the abundance of melt/fluids trapped in the adjacent mantle (Pilet *et al.*, 2011). In locations that experienced Pan-African-aged MARID (mica–amphibole–rutile–ilmenite–diopside) metsomatism, mafic volcanic products commonly display unradiogenic Nd and Hf isotopic ratios (Tappe *et al.*, 2008), because minor metasomatic pyroxene and Ti-oxide can significantly affect time-integrated isotopic signatures by decreasing the Lu/Hf and Sm/Nd ratios (Pearson & Nowell, 2004; Choukroun *et al.*, 2005). Consequently, lavas derived from ancient amphibole–pyroxene–Ti-oxide metasomes should display unradiogenc ε_{Hf} and thus plot below the Hf–Nd mantle array. Lavas from Gerba Guracha (Rooney *et al.*, 2014) and Toro Ankole (Uganda; Rosenthal *et al.*, 2009) were clearly derived from metasomatized mantle and do indeed plot below the mantle array (Fig. 9). Similarly, Miocene lavas from Turkana, Bale and Debre Sahil all have ε_{Hf} values below the mantle array, forming a trend towards EMI compositions, attributed to melts derived from the metasomatized Pan-African lithosphere. Volcanic suites that lack a HIMU Pb isotopic signature (Chiro, Pliocene and Quaternary Bale) plot close to or above the Hf–Nd mantle array, demonstrating the absence of ancient metasomatism and requiring metasomatic activity in these regions to be of recent origin.

A strength of this model is its ability to explain the timing of episodic volcanism and topographic uplift across the region. The Ethiopian Plateau has experienced ~2 km of uplift since the Oligocene (Gani *et al.*, 2007). Pik *et al.* (2006) suggest uplift occurred shortly after the eruption of Oligocene flood basalts, while other models invoke multiple stages of uplift over the past 30 Ma, with significant uplift occurring since the Miocene (Weissel *et al.*, 1995; Gani *et al.*, 2007). Gani *et al.* (2007) suggest that uplift occurred in three phases (29–10 Ma, 10–6 Ma and 6 Ma–present), the most recent of which is marked by an almost 200% increase in uplift rate from 120 m/my to 320 m/my. This significant and abrupt uplift rate increase at 6 Ma led Gani *et al.* (2007) to suggest that erosional unloading via lithospheric foundering may play a critical role in adjusting lithospheric buoyancy. As the HIMU component does not appear in any lavas younger than 10 Ma, the metasomatized lithosphere containing this component may have been removed during one or more local or regional lithospheric removal events (Furman *et al.*, 2016). Pliocene and Quaternary Bale lavas were generated at shallower depths and are isotopically distinct from their Miocene counterparts, consistent with removal of ancient lithosphere.

Model 2: Heterogeneous plume contributions to Ethiopian Plateau volcanism

The deep, long-lived African superplume is thermally and geochemically heterogeneous and may ultimately be the source for chemical heterogeneities recorded throughout the EARS. Entrained, metasomatically

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altered, recycled oceanic lithosphere within the deep mantle would provide a long-term, low $^{3}He/^{4}He$ chemical reservoir capable of contributing to magmatism and metasomatism over a large area and could persist through much of EARS evolution. Several geochemical studies suggest that the HIMU signature is contained in easily fusible, small-scale, chemically distinct blebs or ribbons of material within an upwelling mantle plume (e.g. Furman *et al.*, 2006b; Meshesha & Shinjo, 2008; Nelson *et al.*, 2012; Natali *et al.*, 2016). Other models suggest the plume itself is geochemically zoned (Beccaluva *et al.*, 2009; Rooney *et al.*, 2018). Geophysical studies support a compositionally heterogeneous African Superplume (Ni *et al.*, 2002; Tan & Gurnis, 2005; 2007; Simmons *et al.*, 2007; Sun & Miller, 2013). This model is attractive in that it can explain the existence of this isotopic signature in early Miocene lavas from Debre Sahil, Bale Mountains and Turkana but its absence from the younger lavas in the same regions.

The heterogeneous plume model is appealing because plume heterogeneities have been invoked elsewhere in the world (e.g. Abouchami *et al.*, 2005; Huang *et al.*, 2011) and some spatial problems associated with the lithosphere model are better resolved when viewed in light of a heterogeneous plume. If the HIMU signature indeed resides in the lithosphere, it must define a layer that survived extension along the Anza graben (Ebinger & Ibrahim, 1994) some 100–150 Ma prior to the modern rifting episode. The HIMU component must also be present and available for melting beneath much of the Ethiopian Plateau, which did not undergo Mesozoic extension. Neither the 30 Ma flood basalts in Ethiopia and Yemen nor the Oligocene–Miocene Chiro basalts show obvious HIMU Pb isotopic signatures. Even after thermal erosion of the base of the lithosphere by the Afar plume, the Ethiopian crust and lithosphere (35–45 km and 70–80 km, respectively; Dugda *et al.*, 2007) are thick compared to estimates of the pre-thinned Turkana region. Rifting did not progress through Ethiopia until after ~20 Ma, i.e. after emplacement of the Miocene Bale and Debre Sahil lavas. These spatial and temporal requirements for a HIMU-trending lithospheric domain appear confounded by the physical differences in lithospheric thickness across the study region. Instead, these compositional variations can be explained by the presence of geochemical heterogeneities in the upwelling mantle plume.

This model cannot, however, readily explain HIMU contributions to lavas in the broader Afro-Arabia region where there is not strong evidence for mantle plume contributions. HIMU intraplate basalts in Sudan, southern Egypt and the western Arabian Peninsula (Bertrand *et al.*, 2003; Lucassen *et al.*, 2008) have been attributed to melts of metasomatized lithosphere. Peridotite xenoliths in Jordan also record a HIMU signature attributed to Pan-African-aged metasomatism rather than recent interaction with the Afar plume (Shaw *et al.*, 2007).

Of these two models, the first explains the presence of heterogeneous metasomatic signatures across the EARS including those regions that do not appear to have significant mantle plume contributions. The broad magmatic evolution of volcanism across the Ethiopian Plateau can be summarized in light of this model.

Geodynamic evolution of southeastern Ethiopian Plateau volcanism

Basaltic lavas on the southeastern Ethiopian Plateau have spatial and temporal geochemical trends that require distinct mineralogical and isotopic source domains. Key stages in the geodynamic evolution of this portion of the EARS, with a focus on the Chiro, Debre Sahil and Bale Mountains regions, include the following and are illustrated in Fig. 14:

1. 45–34 Ma: Small volume basaltic magmas were produced in southern Ethiopia and Turkana (George *et al.*, 1998, George & Rogers, 2002; Furman *et al.*, 2006b). The southern Ethiopia lavas were likely derived by partial melting of thickened, metasomatized lithospheric mantle, leading to more radiogenic $^{87}Sr/^{86}Sr$ ($>0{\cdot}7037$) and less radiogenic $^{143}Nd/^{144}Nd$ ($<0{\cdot}5127$) and $^{206}Pb/^{204}Pb$ ($<19{\cdot}1$) relative to Afar plume compositions (Rooney *et al.*, 2012). Coeval lavas from Turkana were HIMU-trending and dominantly derived from metasomatized lithospheric mantle, but their major and trace element characteristics suggest they also experienced significant crustal contamination, making it difficult to identify the true nature of mantle contributions (George & Rogers, 2002; Furman *et al.*, 2006b).
2. 30 ± 2 Ma: The high $^{3}He/^{4}He$ Afar plume produced voluminous flood basalts in northern Ethiopia, including the earliest lavas in Chiro. Flood basalts and picrites record contributions from both the plume and metasomatized lithospheric mantle (Pik *et al.*, 1999; Beccaluva *et al.*, 2009; Furman *et al.*, 2016; Natali *et al.*, 2016). During this time, lithospheric erosion (e.g. thermal erosion, lithospheric drips, etc.) consumed much of the metasomatic material beneath the flood basalt province (30–50 km; Dugda *et al.*, 2007).
3. 28–11 Ma: Shield volcanoes in northwest Ethiopia formed on top of flood basalt sequences and have compositions that range from Afar plume-like to HIMU-trending (Choke and Guguftu volcanoes, Bure volcanic field, Gerba Guracha volcanic field; Kieffer *et al.*, 2004; Meshesha & Shinjo, 2007; Rooney *et al.*, 2014, 2017b). Basalts erupted in northeast Ethiopia (Chiro, 19–15 Ma) lack a HIMU character and resemble dominantly anhydrous lithospheric mantle melts. This suggests that the ancient HIMU-like metasomatic component was either never present or had been removed from beneath northeast Ethiopia. Thus far, it is unclear if there was any involvement of the Afar plume in the regional

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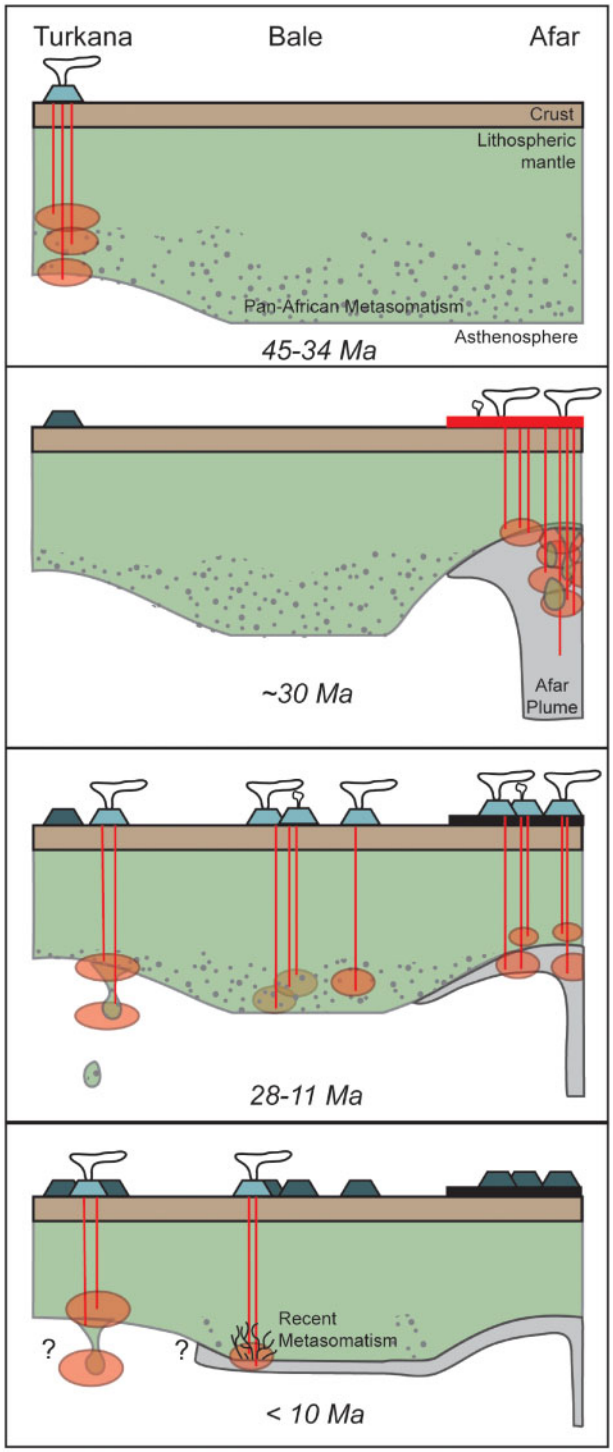


Fig. 14. Cross-sectional cartoon of volcanic episodes on the southeast Ethiopian Plateau from the Afar depression, through the Bale Mountains, to the Turkana Depression. The major events characterizing each time period are outlined in the discussion.

magmatism during this time. The Miocene Chiro lavas do not resemble the depleted upper mantle isotopically, but the $^{3}He/^{4}He$ value of 9 R_A lies in the upper portion of the range of mid-ocean ridge basalts (~7–10 R_A; Graham *et al.*, 2014). Conversely, this may also be evidence for a weak volatile contribution of the Afar plume to the dominantly lithospheric mantle melts that were being erupted at this time.

4. To the south, early volcanism in Debre Sahil (24 Ma) and the Bale Mountains (26–25 Ma) produced mafic alkaline lavas and silicic tuffs. The basaltic lavas were derived predominantly from lithospheric mantle with an ancient yet variable metasomatic history. Transitional Miocene Bale lavas originated as partial melts within the garnet stability field, deeper than the earlier alkaline Bale lavas. Thus there was a progression from melting of an amphibole-dominated mineral assemblage to melting a deeper, more anhydrous assemblage. This change may have resulted from: (1) initial melting of shallow amphibole metasomes, followed by melting of deeper pyroxenitic veins; or (2) localized drips of metasomatised lithosphere descending into the hotter asthenosphere; the drips melted incrementally, producing early hydrous melts followed by increasingly anhydrous types (Furman *et al.*, 2016). Additionally, the Turkana region experienced renewed volcanism (26-17 Ma) with a strong HIMU signature (Furman *et al.*, 2006b). These lavas may have been produced by melting lithospheric drips of anciently metasomatized lithospheric mantle (Furman *et al.*, 2016).
5. 10 Ma–Recent: Localized lithospheric removal led to increased plateau uplift, volcanism and rift propagation throughout the northern EARS. The Turkana region experienced renewed volcanism from the mid-Pliocene through the Quaternary (Furman *et al.*, 2004, 2006b). This young episode of volcanism lacks the pronounced HIMU character seen in the preceding Miocene lavas, instead more closely resembling the geochemistry of Huri Hills lavas (Class *et al.*, 1994). In Ethiopia, Pliocene and Quaternary Bale lavas record contributions from the Afar plume in addition to those from the shallow lithospheric mantle. The shallow lithospheric mantle source had been recently metasomatized (post-Miocene, possibly in relation to the Afar plume). The Pliocene Bale lavas originated as low-degree melts derived dominantly from amphibole-rich material. Quaternary Bale lavas originated as larger degree melts, following the depletion of the metasomatic assemblage caused by the earlier Pliocene melting. A lessening contribution from this lithospheric mantle source, mixed with Afar plume material, generated the young lavas that have mildly elevated $^{3}He/^{4}He$ (~12 R_A), similar to Quaternary lavas from the MER (12–15 R_A; Rooney *et al.*, 2012).

CONCLUSIONS

Lavas emplaced on the southeastern Ethiopian Plateau record spatially and temporally variable contributions from several magma sources. In the north,

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Oligocene–Miocene lavas from Chiro are geochemically and isotopically similar to 30 Ma HT1 basalts derived from the Afar plume and the lithospheric mantle, especially with respect to their elevated TiO_2 contents, trace element abundances and isotopic ratios. This observation demonstrates that the regional high-Ti compositional zoning reported on the western plateau can be extended to the southeastern Ethiopian Plateau and includes lavas younger than the flood basalts. To the south, alkaline Miocene basalts (26–24 Ma) from Debre Sahil and the Bale Mountains were produced by partial melting of an amphibole + Ti-oxide ± clinopyroxene-bearing metasomatized lithospheric mantle. The alkaline lavas trend isotopically towards the HIMU geochemical endmember, which may be the time-integrated result of anciently metasomatized lithospheric mantle. In contrast, slightly younger transitional Miocene Bale lavas do not show pronounced HIMU contributions; their trace element signatures are indicative of melting at greater depths. This geochemical shift may be caused by melting deeper, amphibole-poor, pyroxene-rich lithosphere over time. Isotopic data indicate that the Pliocene and Quaternary Bale lavas were derived from a mantle source distinct from that of the underlying older Miocene lavas. Similar to Miocene alkaline Bale lavas, trace element data indicate that Pliocene and Quaternary Bale lavas were also derived from an amphibole-bearing metasomatized mantle source. Pliocene Bale lavas are more enriched in incompatible elements and may represent a smaller degree of melting compared to the subsequent Quaternary Bale lavas. Quaternary lavas were generated at a shallower depth than the Miocene lavas, suggesting the removal of a portion of the lower lithosphere prior to the young volcanic activity. The metasomatic mantle material may have resided in a shallower level of the lithospheric mantle. However, isotopic data for these young lavas lack the HIMU contribution seen in Miocene Bale lavas. Instead, the Pliocene and Quaternary Bale lavas have Sr–Nd–Pb–Hf–He isotopic ratios akin to MER lavas, consistent with contributions from the Afar plume or metasomatism of the shallow lithospheric mantle by the Afar plume.

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SUPPLEMENTARY DATA

Supplementary data are available at *Journal of Petrology* online.

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